

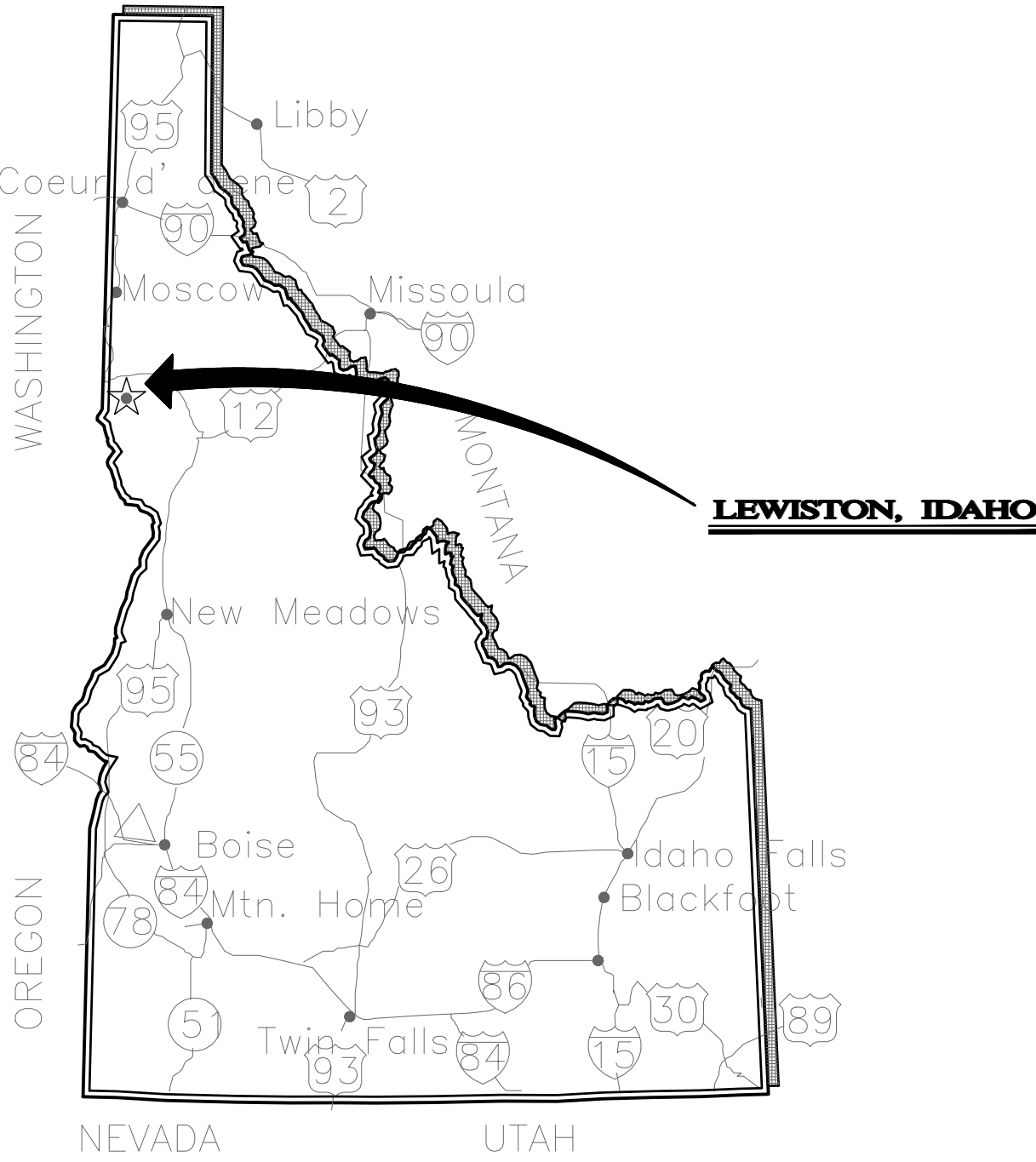
FM22701 REBID

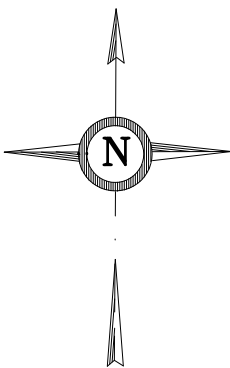
IDAHO TRANSPORTATION DEPARTMENT

HEATING SYSTEM UPGRADES

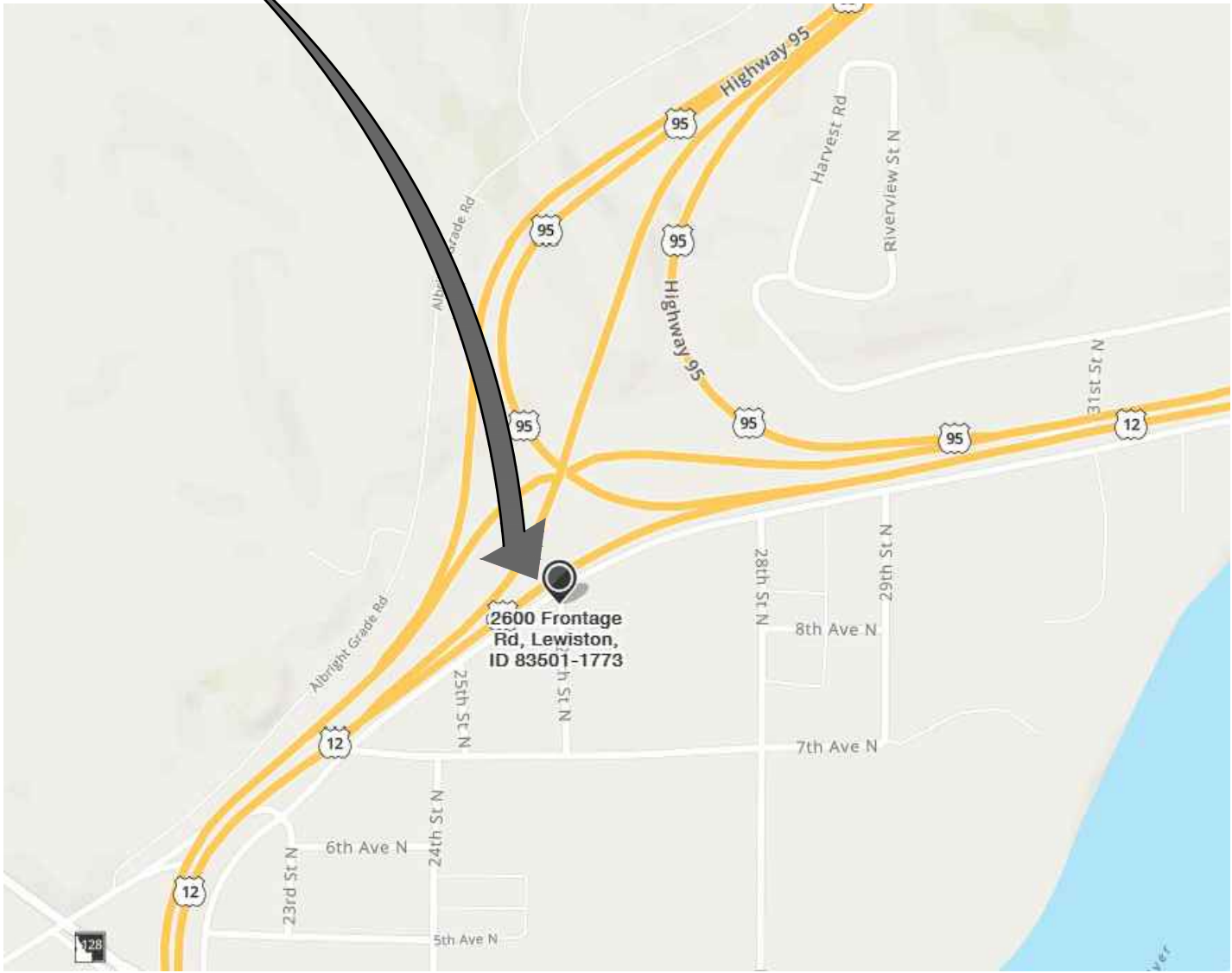
DISTRICT 2

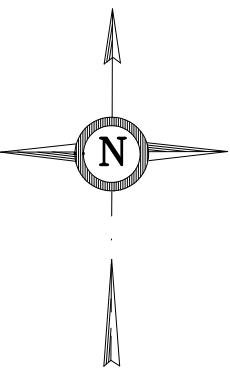
2606 Frontage Rd.
Lewiston, Idaho 83501



 **State Map**
no scale

PROJECT LOCATION
IDAHO TRANSPORTATION DEPARTMENT
LEWISTON, IDAHO

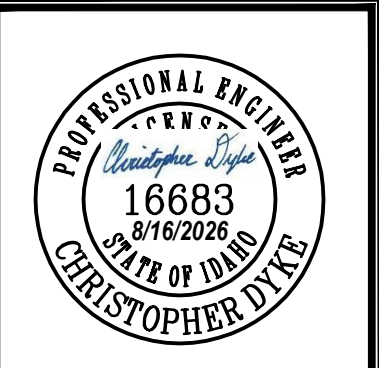


 **Lewiston, Idaho**
Partial City Vicinity Map
no scale

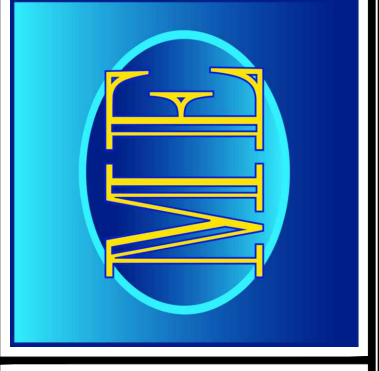
INDEX OF DRAWINGS

SHEET	DESCRIPTION
G0.0	General Cover Sheet
M0.0	Mechanical Cover Sheet
M1.0	Mechanical Floor Plans
M2.0	Mechanical Details
M3.0	Mechanical Schedules
M4.0	Mechanical Controls
M4.1	Mechanical Controls
E0.0	Electrical Cover Sheet
E1.0	Mechanical Power Plan

NO.	REVISIONS	DATE



MUSGROVE
ENGINEERING, P.A.
234 S. Whisperwood Way
Boise, Idaho 83709
208-384-0585
www.musgrovepa.com
OVER 50 YEARS OF EXCELLENCE



REBID - ITD DISTRICT 2
HEATING SYSTEM UPGRADES
2606 FRONTAGE RD
LEWISTON, ID 83501

PROJECT	26-150
DRAWN	JD
CHECKED	CD
DATE	AUGUST 2026
SHEET NAME	GENERAL COVER SHEET
G0.0	

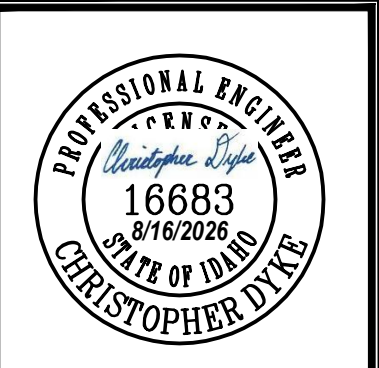
MECHANICAL ABBREVIATIONS			
A/C or AC	AIR CONDITIONING	KW	KILOWATT
AF	ABOVE FINISHED FLOOR	KWH	KILOWATT HOUR
AHU	AIR HANDLING UNIT		
ASHRAE	AMERICAN SOCIETY OF HEATING, REFRIGERATION, AND AIR CONDITIONING ENGINEERS	LAT	LEAVING AIR TEMPERATURE
		LAV	LAVATORY
BTU	BRITISH THERMAL UNITS	LEED	LEADERSHIP IN ENERGY AND ENVIRONMENTAL DESIGN
BTUH	BTUS PER HOUR	LWT	LEAVING WATER TEMPERATURE
		MAX	MAXIMUM
CA	COMBUSTION AIR	MCA	MINIMUM CIRCUIT AMPS
CC	COOLING COIL	MOC	MAXIMUM OVERCURRENT PROTECTION
CFM	AIR FLOW RATE (CUBIC FEET PER MINUTE)	MIN	MINIMUM
CHWR	CHILLED WATER RETURN	NC	NOISE CRITERIA
CHWS	CHILLED WATER SUPPLY	NFPA	NATIONAL FIRE PROTECTION ASSOCIATION
CLS	CEILING	NTS	NOT TO SCALE
CW	COLD WATER	OSA	OUTSIDE AIR
		PD	PRESSURE DROP
DEG or °	DEGREE	PH or Ø	PHASE
DIA or Ø	DIAMETER	PRV	PRESSURE REDUCING VALVE
DB	DRY BULB		
EA	EXHAUST AIR		
EAT	ENTERING AIR TEMPERATURE		
EER	ENERGY EFFICIENCY RATIO		
ESP	EXTERNAL STATIC PRESSURE	RA	RETURN AIR
EWI	ENTERING WATER TEMPERATURE	RPM	REVOLUTIONS PER MINUTE
		RTU	ROOFTOP UNIT
FCO	FLOOR CLEANOUT	SA	SUPPLY AIR
FD	FIRE DAMPER	SEER	SEASONAL ENERGY EFFICIENCY RATIO
FLA	FULL LOAD AMPS	SFD	COMBINATION SMOKE/FIRE DAMPER
FLR	FLOOR	SP	STATIC PRESSURE
PPM	FEET PER MINUTE	SYM	SYMBOL
FT	FEET		
		T & P	TEMPERATURE AND PRESSURE
GA	GALUIGE	TEMP	TEMPERATURE
GCO	GRADE CLEANOUT	TYP	TYPICAL
GPM	WATER FLOW RATE (GALLONS PER MINUTE)		
		UMC	UNIFORM MECHANICAL CODE
HC	HEATING COIL	UPC	UNIFORM PLUMBING CODE
HP	HORSE POWER	URL	URINAL
HVAC	HEATING, VENTILATING, AIR CONDITIONING		
HW	HOT WATER		
HWR	HOT WATER RETURN	VTR	VENT THROUGH ROOF
HWS	HOT WATER SUPPLY	V	VOLTS
		W/	WITH
IBC	INTERNATIONAL BUILDING CODE	WB	WET BULB
IECC	INTERNATIONAL ENERGY CONSERVATION CODE	WC	WATER CLOSET
IFC	INTERNATIONAL FIRE CODE	WCO	WALL CLEANOUT
IFGC	INTERNATIONAL FUEL GAS CODE	WH	WATER HEATER
IMC	INTERNATIONAL MECHANICAL CODE		
IPC	INTERNATIONAL PLUMBING CODE		
NOTE: THIS IS A STANDARD LIST OF COMMONLY USED MECHANICAL ABBREVIATIONS. SOME OF THE ABBREVIATIONS SHOWN ABOVE MAY NOT BE USED IN THIS DRAWING PACKAGE.			

MECHANICAL GENERAL NOTES	
1.	ALL MECHANICAL EQUIPMENT AND SYSTEMS SHALL BE INSTALLED IN ACCORDANCE WITH THE LATEST ADOPTED MECHANICAL CODE, AND ALL LOCAL & STATE CODES.
2.	ALL PLUMBING EQUIPMENT AND SYSTEMS SHALL BE INSTALLED IN ACCORDANCE WITH THE LATEST ADOPTED PLUMBING CODE, AND ALL LOCAL & STATE CODES.
3.	ALL MECHANICAL AND PLUMBING EQUIPMENT SHALL BE INSTALLED PER THE MANUFACTURER'S RECOMMENDATIONS.
4.	MECHANICAL CONTRACTORS SHALL RECEIVE PRIOR APPROVAL FROM THE STRUCTURAL ENGINEER BEFORE MAKING CUTS THROUGH ANY STRUCTURAL MEMBER.
5.	MECHANICAL CONTRACTORS SHALL COORDINATE INSTALLATION WITH CONSTRUCTION SUPERVISOR AND WITH ALL OTHER TRADES TO AVOID CONFLICTS.
6.	THE MECHANICAL CONTRACTORS SHALL VERIFY MOTOR VOLTAGES WITH THE ELECTRICAL DRAWINGS BEFORE ORDERING MOTORIZED EQUIPMENT AND CONTROLS.
7.	SEE MECHANICAL SCHEDULE SHEET FOR SCHEDULED CAPACITIES OF ALL MECHANICAL EQUIPMENT AND MATERIALS SPECIFIED.
8.	ALL MECHANICAL EQUIPMENT TO BE PROPOSED MUST BE ON THE APPROVED LIST PRIOR TO SUBMITTALS. ALL APPROVED MANUFACTURERS MUST BE CAPABLE OF MEETING THE REQUIREMENTS OF THE SPECIFIED EQUIPMENT.
9.	RUNOUT AND HOOKUP SIZES TO INDIVIDUAL PLUMBING FIXTURE CAN BE FOUND ON THE PLUMBING FIXTURE SCHEDULE.
10.	PROVIDE REMOTE CEILING ACCESS BALANCE DAMPERS WITH CONCEALED CHROME PLATE COVERS FOR BALANCE DAMPERS LOCATED ABOVE HARD CEILINGS.
11.	PAIN ALL VTR'S, FLUES, EXHAUST CAPS, AND OTHER MECHANICAL ITEMS ON THE ROOF TO MATCH THE ROOF COLOR.
12.	INSULATED FLEXIBLE DUCTWORK MAY BE USED FOR RUNOUTS TO GRILLES AND DIFFUSERS, IN LENGTHS OF 6'-0" OR LESS.
13.	MAINTAIN MINIMUM OF 10'-0" DISTANCE BETWEEN ALL FRESH AIR INTAKES AND EXHAUST OR GAS FLUE DISCHARGES.
14.	LOCATE ACCESS HATCHES SO AS TO PROVIDE OPTIMUM SERVICEABILITY TO EQUIPMENT AND/OR VALVING. SEE ARCHITECTURAL SPECIFICATION FOR TYPE AND COLOR. COORDINATE LOCATION WITH STRUCTURAL & LIGHTING.
15.	WHENEVER THERE IS A DISCREPANCY BETWEEN THE RUNOUT DUCT SIZE SHOWN ON THE PLANS AND THAT SHOWN IN THE SCHEDULE, ALWAYS USE THE LARGER OF THE TWO DUCT SIZES.
16.	PROVIDE REQUIRED TESTING, ADJUSTING, AND BALANCING IN ACCORDANCE WITH 107.2.5.3 2018 IMC, REPORTS TO BE PROVIDED TO CODE OFFICIAL PRIOR TO MECHANICAL FINAL INSPECTION, PROVIDED BY THIRD PARTY QUALIFIED AGENCY.
17.	THE CONTRACTOR SHALL BE HELD RESPONSIBLE FOR VERIFICATION OF EXISTING JOB CONDITIONS PRIOR TO BID. NO ADDITIONAL COST SHALL BE AWARDED TO THE SUCCESSFUL CONTRACTOR (OR THEIR SUBCONTRACTORS) AFTER BIDS HAVE BEEN SUBMITTED AND CONTRACTS AWARDED FOR FAILURE TO VERIFY EXISTING FIELD CONDITIONS. DISCREPANCIES BETWEEN ACTUAL FIELD CONDITIONS AND CONTRACT DOCUMENTS SHALL BE BROUGHT TO THE ENGINEERS ATTENTION FOR ALTERNATIVE METHODS OF INSTALLATION PRIOR TO THE BIDDING OF THIS PROJECT.
18.	UNLESS OTHERWISE NOTED ALL EXISTING MECHANICAL EQUIPMENT, PIPING, ETC. TO BE REMOVED SHALL BE DISPOSED OF BY THE CONTRACTOR UNDER THIS CONTRACT. THE OWNER SHALL RETAIN THE RIGHT TO KEEP ANY REMOVED ITEMS.
19.	HOLES IN EXISTING WALL OR FLOORS SHALL BE PATCHED TO MATCH EXISTING WHERE PIPING, DUCTWORK, ETC. WERE REMOVED OR ADDED DURING THIS PROJECT.
20.	DAMAGE TO THE EXISTING FACILITY DURING THE CONSTRUCTION SHALL BE REPAIRED OR REPLACED BY THE CONTRACTOR AT NO COST TO THE OWNER.

MECHANICAL AND PLUMBING DRAWINGS LEGEND			
	FLEXIBLE DUCTWORK		THREE WAY CONTROL VALVE
	DUCTWORK		TWO WAY CONTROL VALVE
	DUCTWORK BREAK		PRESSURE REDUCING VALVE
	DUCTWORK OR PIPING RISE		GATE VALVE
	CONCENTRIC SQUARE TO ROUND TRANSITION		REDUCER
	MOTORIZED DAMPER		GLOBE VALVE
	MANUAL VOLUME DAMPER		BALL VALVE
	SPIN-IN FITTING W/ AIR EXTRACTOR AND HAND DAMPER		BUTTERFLY VALVE
	HIGH EFFICIENCY FITTING W/ HAND DAMPER		BALANCE VALVE
	SWITCH		CHECK VALVE
	THERMOSTAT		FLOOR CLEANOUT
	HUMIDISTAT		WALL CLEANOUT
	TEMPERATURE SENSOR		GRADE CLEANOUT
	CARBON DIOXIDE SENSOR		WATER HAMMER ARRESTOR
	CARBON MONOXIDE SENSOR		FLOOR DRAIN
	NITROGEN DIOXIDE SENSOR		FLOOR SINK
	DUCT SMOKE DETECTOR		GAS PRESSURE REGULATOR W/ GAS COCK
	COMBINATION SMOKE/FIRE DAMPER		PRESSURE RELIEF VALVE
	FIRE DAMPER		VENT-THROUGH-ROOF
	SMOKE DAMPER		VENT
	EQUIPMENT CALLOUT		SOIL, WASTE, OR SANITARY SEWER
	TURNING VANES		ACID WASTE LINE
	INTAKE OR EXHAUST		ACID VENT LINE
	DIRECTION OF AIRFLOW		STORM DRAIN
	SUPPLY DIFFUSER		ROOF DRAIN LINE
	RETURN GRILLE		OVERFLOW DRAIN LINE
	EXHAUST GRILLE		CONDENSATE DRAIN LINE
	FLOOR GRILLE		DOMESTIC COLD WATER (CW)
	CEILING EXHAUST FAN		DOMESTIC HOT WATER (HW)
	TEMPERATURE GAUGE		DOMESTIC HOT WATER RETURN (HWR)
	PRESSURE GAUGE (LIQUID FILLED W/ ISOLATION VALVE)		TEMPERED WATER (TW)
	TEMPERATURE SENSOR (DUCT OR PIPING)		MEDIUM PRESSURE NATURAL GAS
	FLOW SWITCH		LOW PRESSURE NATURAL GAS
	STAINLESS STEEL BRAIDED FLEX CONNECTION		FIRE SPRINKLER LINE
	ELASTOMETRIC FLEX CONNECTOR		GEO THERMAL WATER SUPPLY
	SUCTION DIFFUSER		GEO THERMAL WATER RETURN
	Y TYPE STRAINER (1 1/2" OR LARGER PROVIDED W/ BLOW DOWN VALVE)		CHILLED WATER SUPPLY
	FLOW DIRECTION		CHILLED WATER RETURN
	DEMOLITION / EQUIPMENT TO BE REMOVED		CONDENSER WATER SUPPLY
	NEW TO EXISTING CONNECTION POINT		CONDENSER WATER RETURN
	EXISTING		HEATING WATER SUPPLY
	FUTURE		HEATING WATER RETURN
	NEW		LIQUID REFRIGERANT LINE
	REDUCED PRESSURE BACKFLOW PREVENTER		SUCTION REFRIGERANT LINE
	DOUBLE CHECK BACKFLOW PREVENTER		SLOPE PIPE IN DIRECTION OF ARROW
	UNION		PIPE ANCHOR
	AIR VENT		PIPE GUIDE
	TRIPLE DUTY VALVE		CAP
NOTE: THIS IS A LIST OF COMMONLY USED MECHANICAL AND PLUMBING SYMBOLS. SOME OF THE SYMBOLS SHOWN ABOVE MAY NOT BE USED IN THIS DRAWING PACKAGE.			

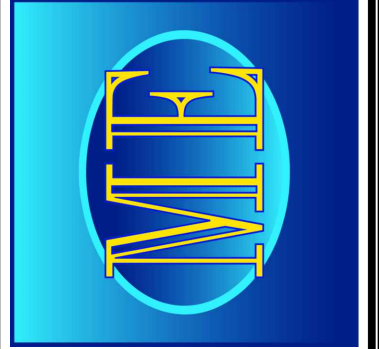
ENERGY CODE COMPLIANCE	
A.	COMPLIANCE WITH THE LATEST ADOPTED EDITION OF THE INTERNATIONAL ENERGY CONSERVATION CODE IS REQUIRED FOR THIS PROJECT. THESE NOTES COVER MANDATORY REQUIREMENTS OF THE CODE. ADDITIONAL REQUIREMENTS ARE NOTED ON THE DRAWINGS AND IN THE SPECIFICATIONS.
B.	MINIMUM REQUIREMENTS FOR SUPPLY AND RETURN DUCTWORK INSULATION: 1. R-6: DUCTS LOCATED IN UNCONDITIONED SPACES (SPACE NEITHER HEATED NOR COOLED SUCH AS ABOVE CEILING SPACES, WALL SPACES, DUCT CHASES, SOFFITS, ATTICS, CRAWL SPACES, UNHEATED BASEMENTS, AND UNHEATED GARAGES). 2. R-12: DUCTS LOCATED OUTSIDE OF THE BUILDING'S INSULATION ENVELOPE (SUCH AS ABOVE THE ATTIC INSULATION). TYPICAL INSULATION THICKNESS REQUIRED TO MEET THESE REQUIREMENTS: 1. FIBERGLASS DUCT WRAP: R-6, R-12. 2. FIBERGLASS DUCT LINER: R-6, R-12.
C.	CONTRACTOR SHALL VERIFY WITH THE MANUFACTURER, THE R-VALUES OF THE ACTUAL INSULATION USED. R-VALUES SHALL BE INSTALLED VALUES.
D.	WHERE DUCTS USED FOR COOLING ARE EXTERNALLY INSULATED, THE INSULATION SHALL BE COVERED WITH A VAPOR RETARDER HAVING A MAXIMUM PERMEANCE OF 0.05 PERM OR ALUMINUM FOIL HAVING A MINIMUM THICKNESS OF 2 MILS. INSULATION HAVING A PERMEANCE OF 0.05 PERMS OR LESS SHALL NOT BE REQUIRED TO BE COVERED. ALL JOINTS AND SEAMS SHALL BE SEALED TO MAINTAIN THE CONTINUITY OF THE VAPOR RETARDER.
E.	ALL DUCT JOINTS, SEAMS, AND CONNECTIONS SHALL BE FASTENED AND SEALED WITH WELDS, GASKETS, ADHESIVES, MASTIC-PLUS-EMBEDDED-FABRIC SYSTEMS, OR TAPES. TAPES AND MASTICS SHALL BE LISTED AND LABELED PER UL181A OR UL181B. DUCT TAPE IS NOT PERMITTED AS A SEALANT ON ANY METAL DUCTS. DUCT CONNECTIONS TO FLANGES OR EQUIPMENT SHALL BE SEALED AND MECHANICALLY FASTENED.
F.	MINIMUM REQUIREMENTS (THICKNESS) FOR PIPING INSULATION: REFER TO PROJECT SPECIFICATIONS.
G.	AN OPERATING AND MAINTENANCE MANUAL SHALL BE PROVIDED PRIOR TO ISSUANCE OF A CERTIFICATE OF OCCUPANCY. THE O&M MANUAL SHALL CONTAIN THE FOLLOWING INFORMATION AS A MINIMUM: 1. EQUIPMENT CAPACITY (INPUT & OUTPUT). 2. EQUIPMENT OPERATING AND MAINTENANCE INSTRUCTIONS. 3. CONTROL SYSTEM MAINTENANCE AND CALIBRATION INFORMATION, INCLUDING WIRING DIAGRAMS, SCHEMATICS, AND CONTROL SEQUENCES. 4. CONTROL SYSTEM SETPOINTS SHALL BE SHOWN ON CONTROL DRAWINGS, AT CONTROL DEVICES, OR IN PROGRAMMING COMMENT ON DDC SYSTEMS. 5. A COMPLETE WRITTEN NARRATIVE ON HOW EACH MECHANICAL SYSTEM IS INTENDED TO OPERATE.

REVISIONS		DATE
NO.		



MUSGROVE
ENGINEERING, P.A.
234 S. Whisperwood Way
Bosc, Idaho 83709
208-384-0585
www.musgrovepa.com
OVER 50 YEARS OF EXCELLENCE

1135 Pier View Dr. #160
Idaho Falls, Idaho 83402
208-523-2862



REBID - ITD DISTRICT 2
HEATING SYSTEM UPGRADES
2606 FRONTAGE RD
LEWISTON, ID 83501

PROJECT	26-150
DRAWN	JD
CHECKED	CD
DATE	AUGUST 2026
SHEET NAME	

MECHANICAL
COVER
SHEET

M0.0

DEMOLITION KEYED NOTES:

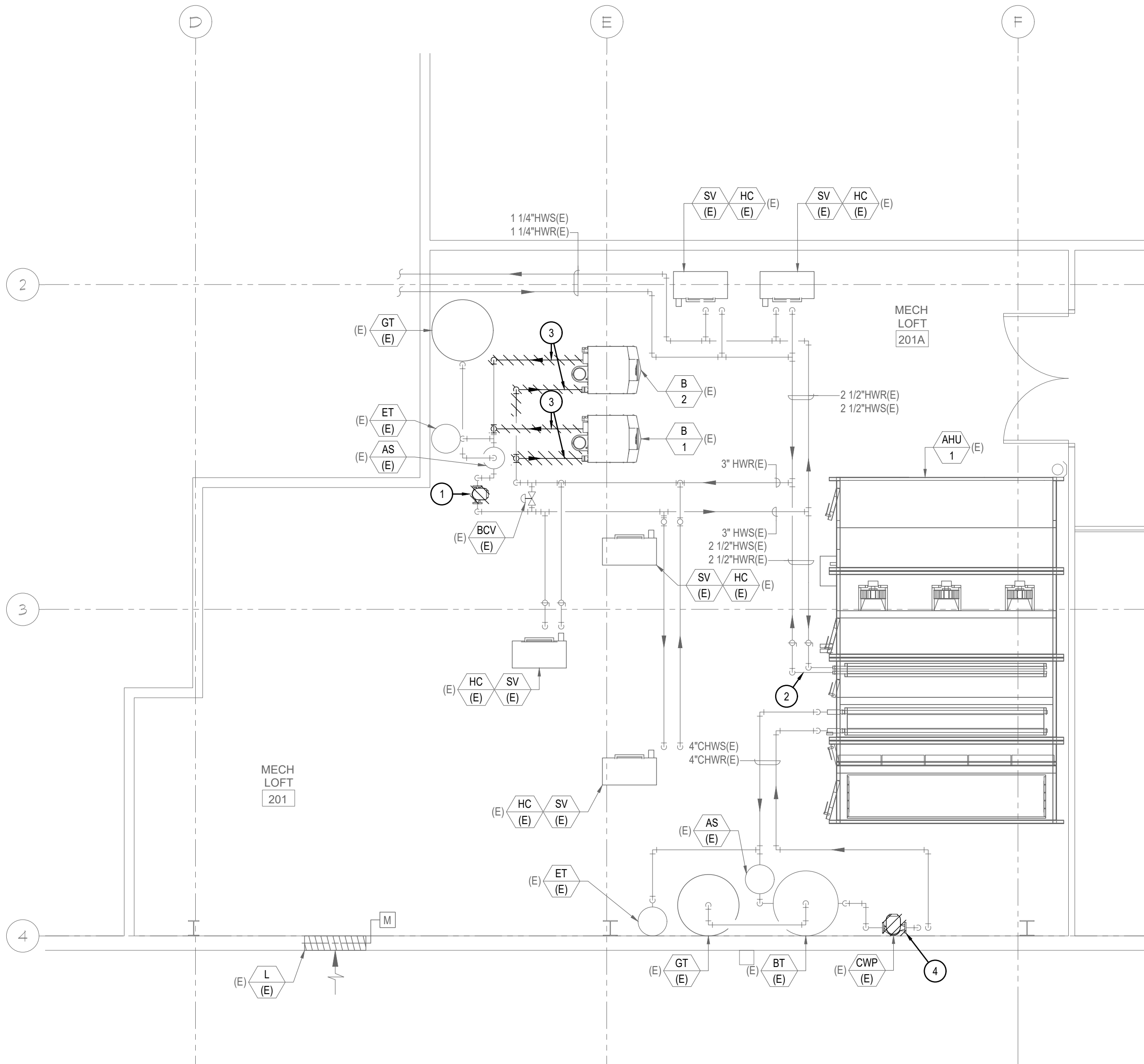
SYMBOL USED FOR NOTE CALLOUT.

1. DISCONNECT AND REMOVE EXISTING HOT WATER PUMPS. EXISTING PIPE AND PIPING ACCESSORIES TO REMAIN AND BE REUSED.
2. DISCONNECT AND REMOVE EXISTING 2-WAY CONTROL VALVE. RETURN VALVE TO OWNER.
3. DISCONNECT AND REMOVE SECTION OF HOT WATER PIPING INCLUDING BOILER MOTORIZED ISOLATION VALVE, STRAINER, CHECK VALVE, AND BALANCE VALVE. SEE NEW WORK FOR CONTINUATION.
4. DISCONNECT AND REMOVE EXISTING CHILLED WATER PUMPS. EXISTING PIPE AND PIPING ACCESSORIES TO REMAIN AND BE REUSED.

KEYED NOTES:

SYMBOL USED FOR NOTE CALLOUT.

1. CLOSE EXISTING BYPASS CONTROL VALVE.
2. CONNECT NEW HOT WATER LOOP PUMPS TO EXISTING PARALLEL PIPING ARRANGEMENT. ADJUST AND MODIFY PIPING AS REQUIRED. SEE DETAIL FOR ADDITIONAL INFORMATION.
3. INSTALL NEW BOILER PUMP ON RETURN SIDE. ADJUST PIPING AS REQUIRED. SEE DETAIL FOR ADDITIONAL INFORMATION.
4. EXTEND EXISTING 3"Ø HOT WATER PIPING AND CONNECT TO INLET OF EXISTING AIR SEPARATOR. SEE PIPING DETAIL FOR ADDITIONAL REQUIREMENTS.
5. INSTALL NEW 3-WAY VALVE ON EXISTING AIR HANDLING UNIT (AHU-1) HEATING WATER CONNECTION. SEE DETAIL FOR ADDITIONAL REQUIREMENTS. BALANCE VALVE TO 48 GPM.
6. MODIFY EXISTING BOILER PROGRAMMING FOR PRIMARY-SECONDARY PIPING ARRANGEMENT. SEE CONTROL DRAWINGS FOR MORE INFORMATION.
7. SEE EXISTING BOILER PIPING DETAIL FOR MODIFICATIONS.
8. CONNECT NEW CHILLED WATER LOOP PUMPS TO EXISTING PARALLEL PIPING ARRANGEMENT. ADJUST AND MODIFY PIPING AS REQUIRED. SEE DETAIL FOR ADDITIONAL INFORMATION.
9. CONNECT NEW 3" DECOUPLED BOILER MANIFOLD TO EXISTING 3" HOT WATER MAIN.



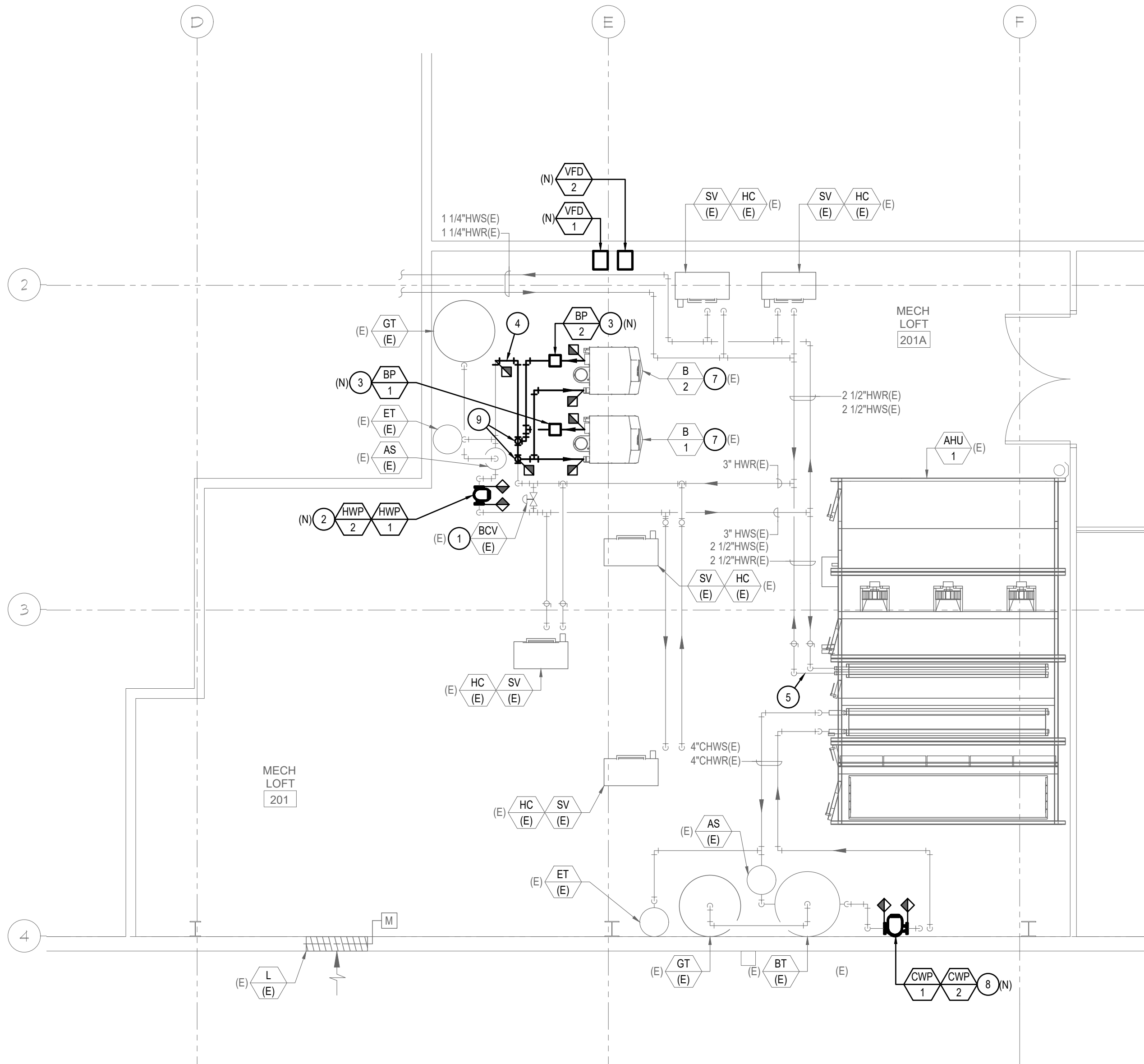
1

LOFT MECHANICAL DEMOLITION PLAN

1/4" = 1'-0"



NORTH



2

LOFT MECHANICAL NEW WORK PLAN

1/4" = 1'-0"

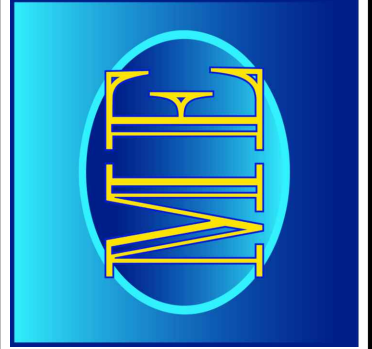


NORTH

NO.	REVISIONS	DATE



MUSGROVE
ENGINEERING, P.A.
1135 Pier View Dr. #160
Bosc, Idaho 83709
208-384-0585
www.musgrovepa.com
OVER 50 YEARS OF EXCELLENCE

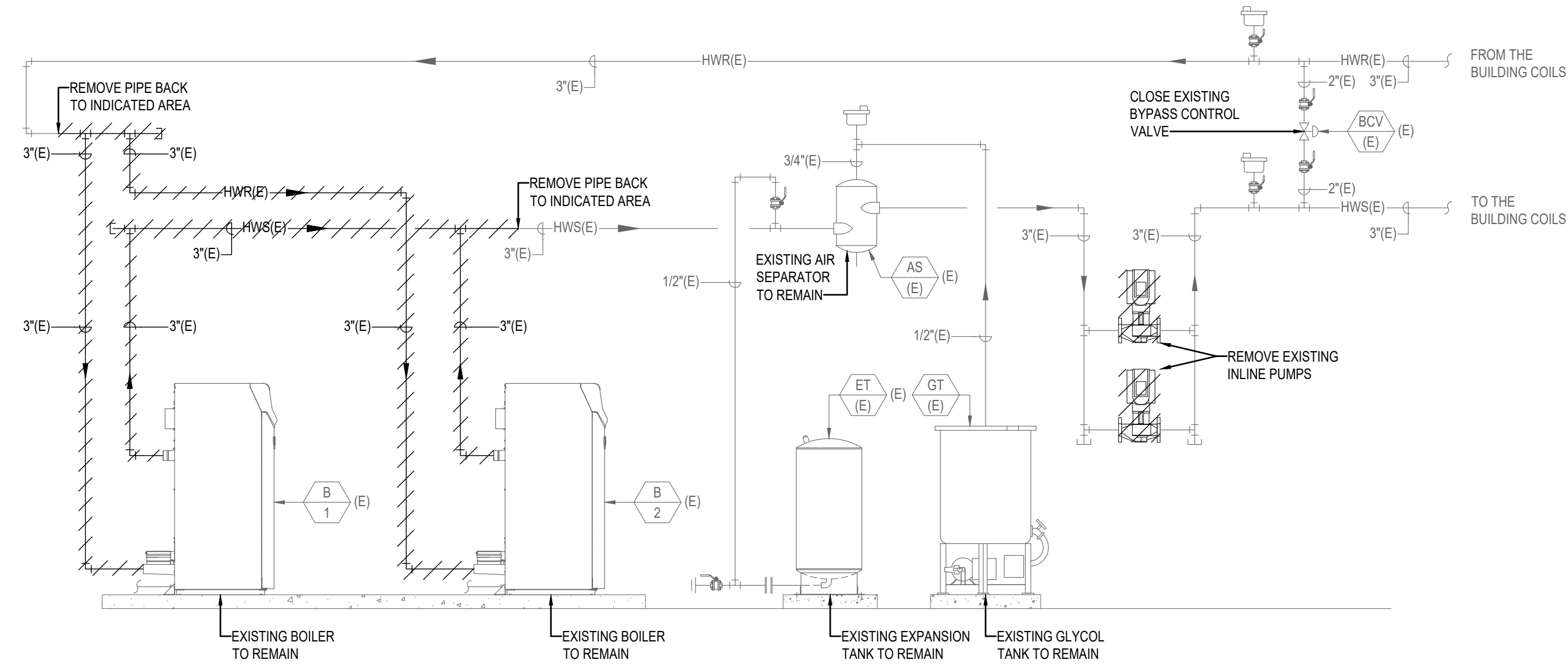


REBID - ITD DISTRICT 2
HEATING SYSTEM UPGRADES
2606 FRONTAGE RD
LEWISTON, ID 83501

PROJECT	26-150
DRAWN	JD
CHECKED	CD
DATE	AUGUST 2026
SHEET NAME	

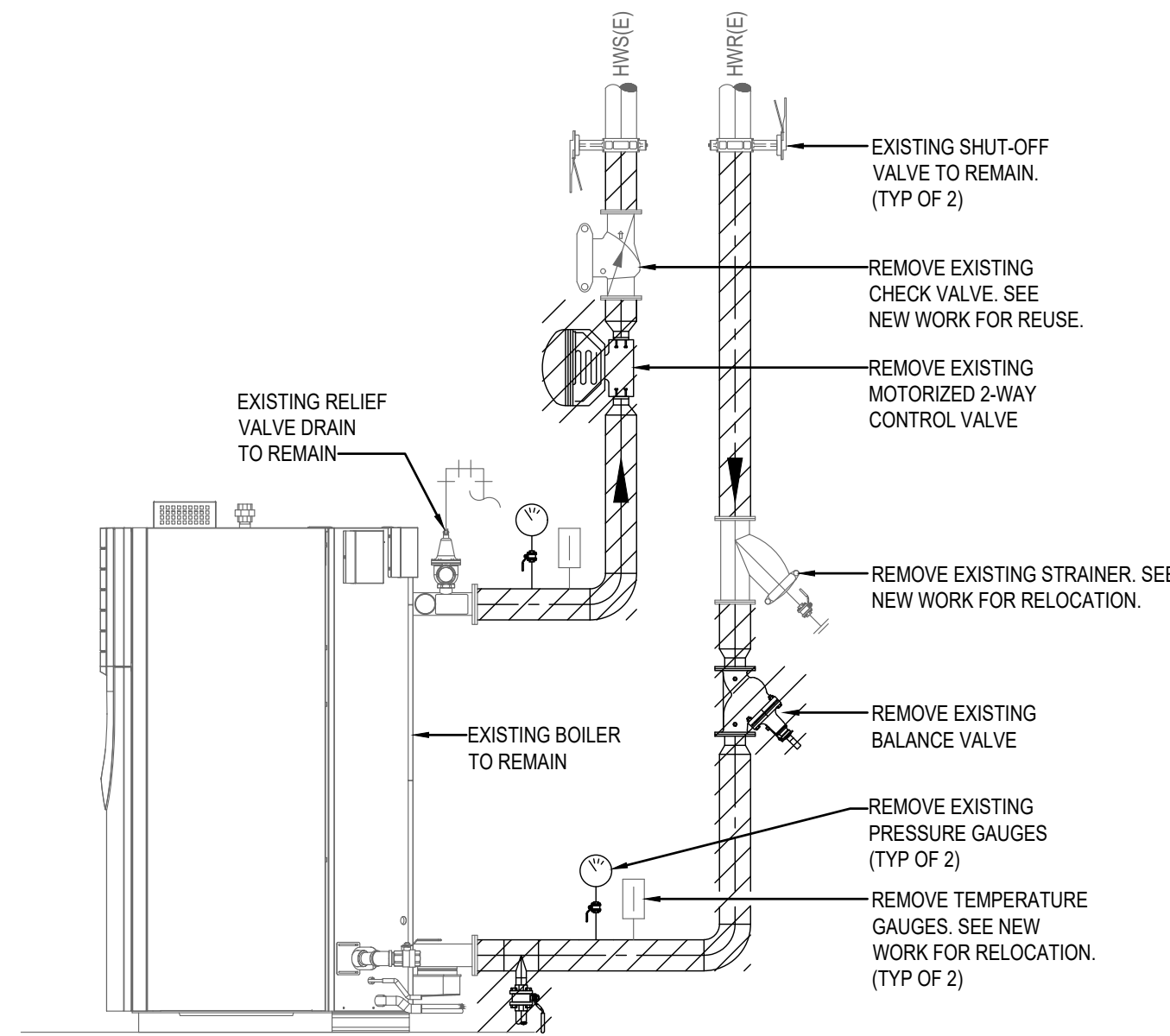
MECHANICAL
FLOOR
PLANS

M1.0



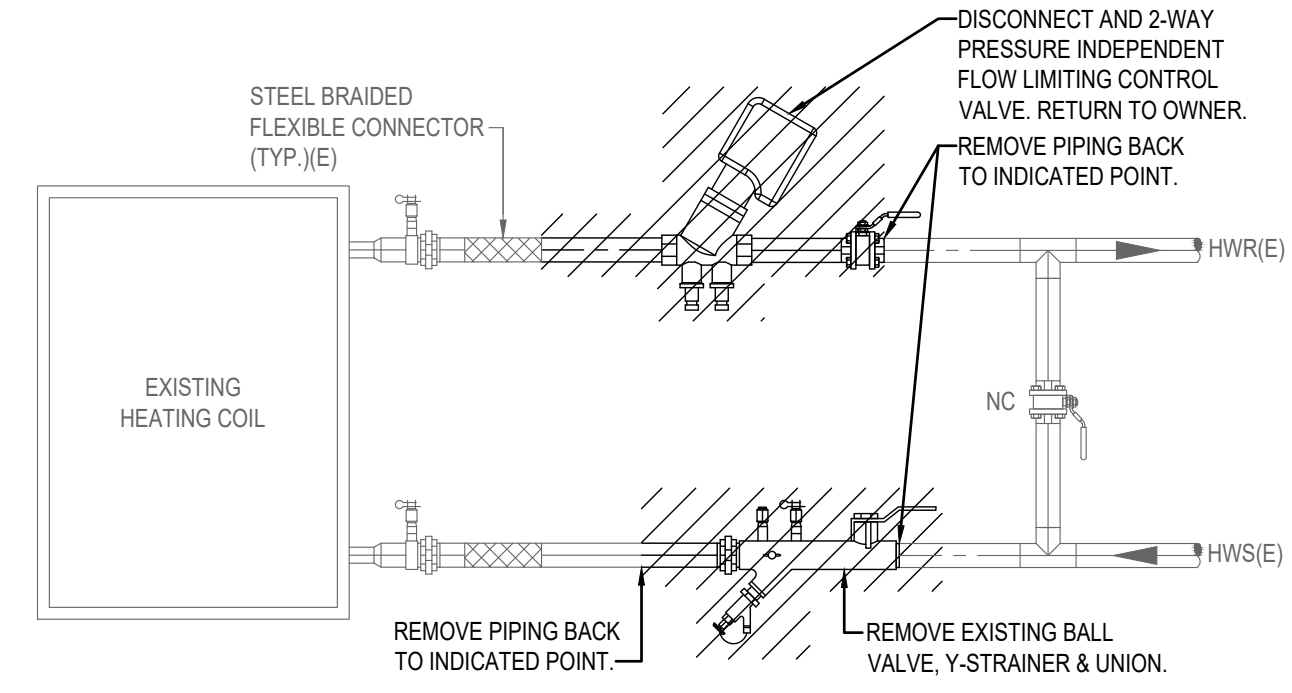
1

HOT WATER SYSTEM DEMOLITION PIPING SCHEMATIC
NOT TO SCALE



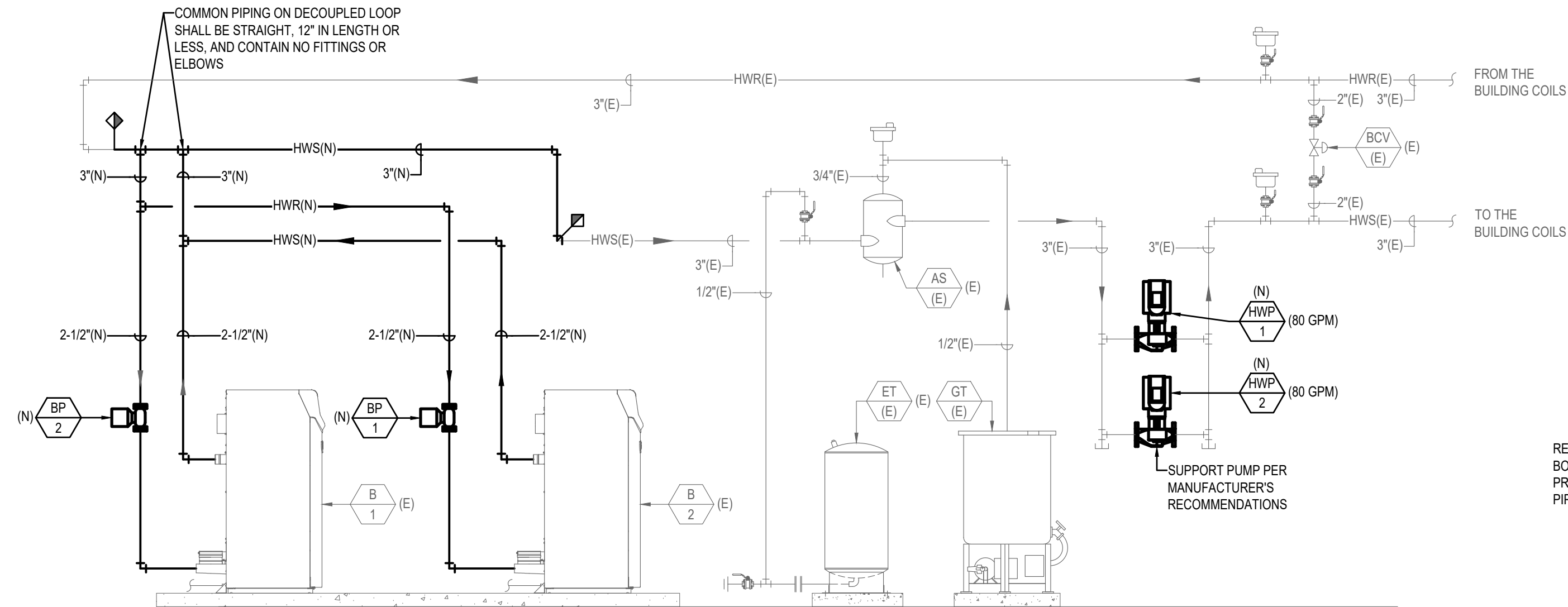
3

BOILER PIPING DEMOLITION DETAIL
NOT TO SCALE



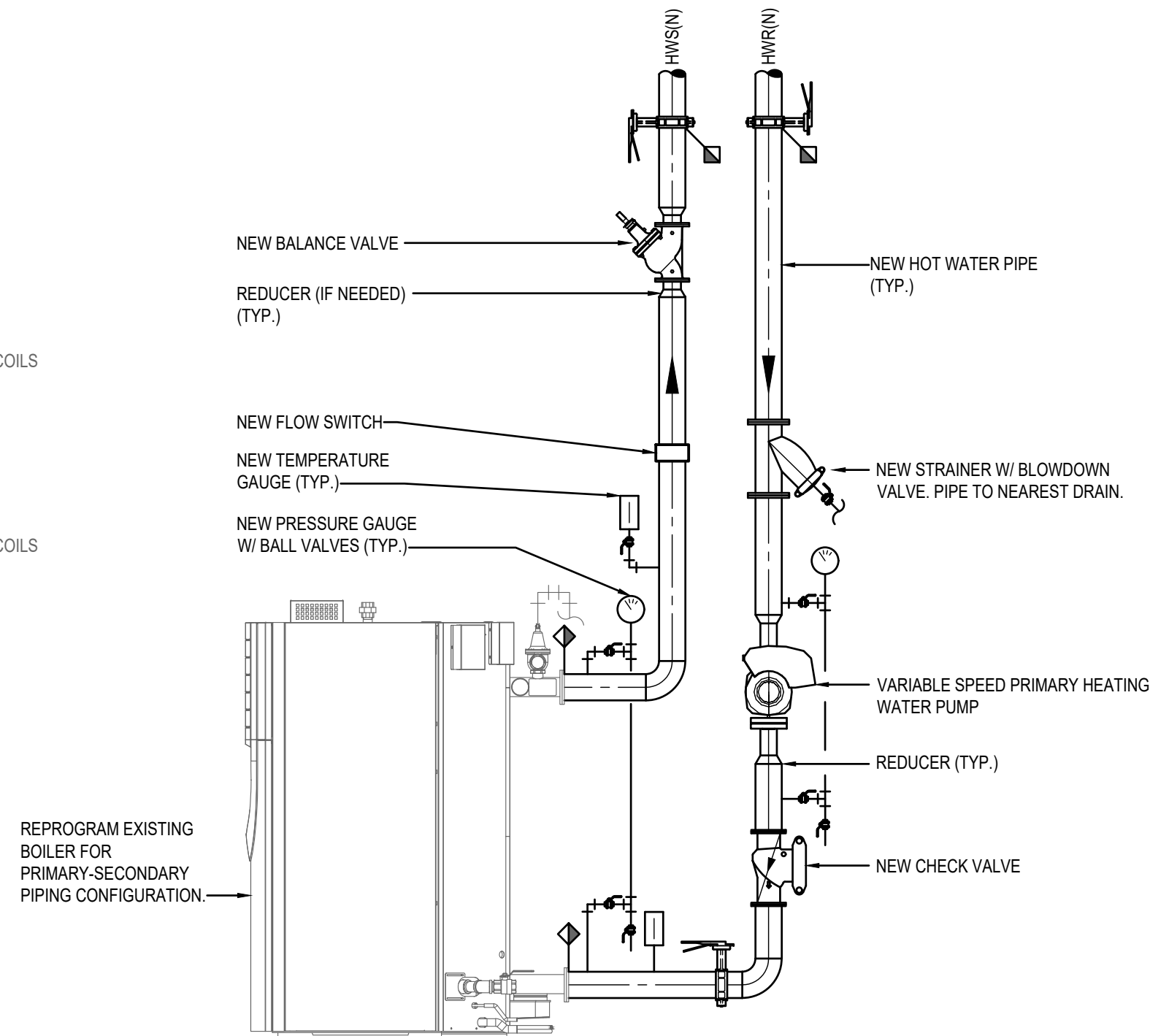
5

EXISTING HEATING COIL PIPING DEMOLITION DETAIL
NOT TO SCALE



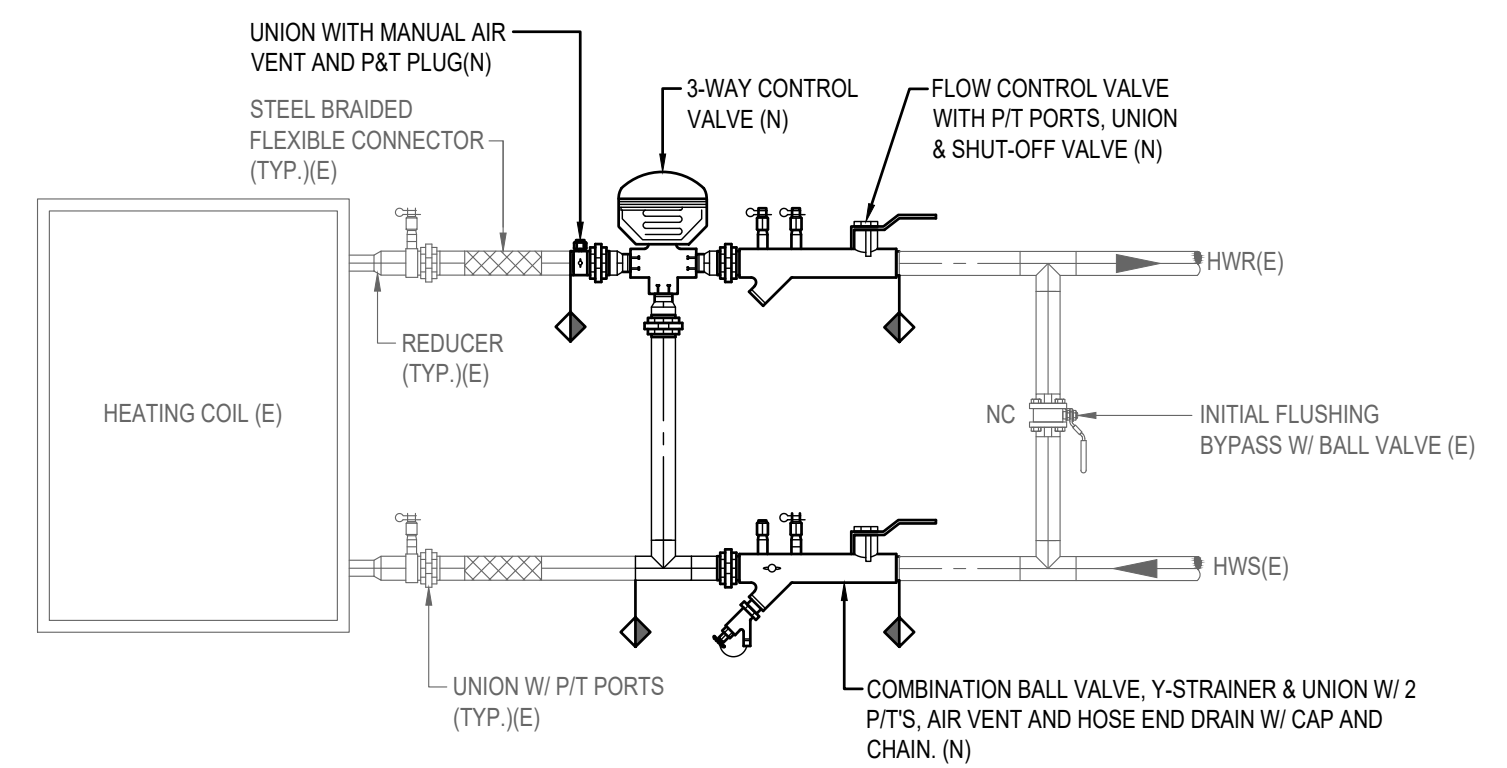
2

HOT WATER SYSTEM NEW WORK PIPING SCHEMATIC
NOT TO SCALE



4

BOILER PIPING NEW WORK DETAIL
NOT TO SCALE



6

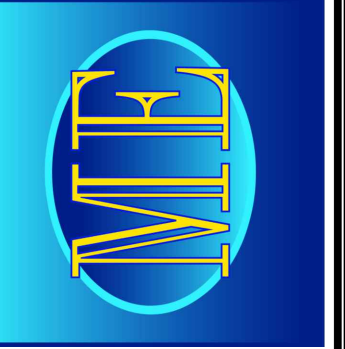
NEW HEATING COIL PIPING DETAIL (3-WAY CONTROL)
NOT TO SCALE

- NOTES:
1. ALL COMPONENTS SHOWN SHALL BE EQUAL TO AUTOFLOW FLOW DESIGN, INC.
 2. INSTALL ALL AIR VENTS, STRAINERS, CONTROL VALVES, BALL VALVES, ETC. IN ACCESSIBLE LOCATIONS FOR MAINTENANCE PURPOSES.

NO.	REVISIONS	DATE



MUSGROVE ENGINEERING, P.A.
1135 Pier View Dr. #160
Bosc, Idaho 83709
208-384-0585
www.musgrovepa.com
OVER 50 YEARS OF EXCELLENCE



REBID - ITD DISTRICT 2
HEATING SYSTEM UPGRADES
2606 FRONTAGE RD
LEWISTON, ID 83501

PROJECT	26-150
DRAWN	JD
CHECKED	CD
DATE	AUGUST 2026
SHEET NAME	

MECHANICAL DETAILS

M2.0

SYMBOL	TYPE	AREA SERVED	SUPPLY FAN							CHILLED WATER/DEHUMIDIFICATION COOLING COIL										HEATING COIL										HUMIDIFICATION COIL				COMBINATION FILTER			MIN. O.S.A. CFM	OPERATING WEIGHT (LBS)	MANUFACTURER AND MODEL	REMARKS									
			TYPE	CFM	ESP	TSP	HP/QTY	V/Ø	RPM	FLUID TYPE	CAPACITY (MBH)		CFM	E.A.T. (°F)		L.A.T. (°F)		MAX FACE VELOCITY (FPM)	AIR P.D. (W.C.)	FLUID PERFORMANCE			FLUID P.D. (FT)	MAX RH	FLUID TYPE	CAPACITY (MBH)	CFM	E.A.T. (°F)	L.A.T. (°F)	MAX FACE VELOCITY (FPM)	AIR P.D. (W.C.)	FLUID PERFORMANCE			FLUID P.D. (FT)	COIL AIR P.D. (FT)					FINNED HEIGHT (IN)	FINNED WIDTH (IN)	FACE AREA (FT²)	FACE VEL. (FPM)	PRE-FILTER		FINAL FILTER		MAX FACE VELOCITY (FPM)
											SENSIBLE	TOTAL		D.B.	W.B.	D.B.	W.B.			E.W.T.	L.W.T.	GPM										E.W.T.	L.W.T.	GPM											EFF.	DEPTH	EFF.	DEPTH	
AHU-1(E)	SEMI-CUSTOM	LABORATORIES	ECM	16,400	2.0	5.1	3.216	460/3	3,169	30% PG	842.7	842.7	16,400	98.7	65.5	50.2	46.9	500	0.8	45	55.0	177.0	18.0	50%	30% PG	960.0	16,400	12.9	66.9	500	0.11	140	98.1	48.0	5.1	0.01	24.0	109.0	36.33	451	MERV 8	2"	MERV 13	4"	500	2,150	7,000	DAIKIN CAH044GDCM	1, 2

1. EXISTING AIR HANDLING UNIT. SPECIFICATIONS SHOWN FOR REFERENCE ONLY.
2. INSTALL NEW 3-WAY VALVE AT HEATING COIL CONNECTION. SEE DETAIL FOR ADDITIONAL REQUIREMENTS.

EXISTING CONDENSING HOT WATER BOILER SCHEDULE											
SYMBOL	AREA SERVED	THERMAL EFFICIENCY	FUEL	EWT (°F)	LWT (°F)	BOILER FLOW (GPM)	MAX P.D. (FT H ₂ O)	CAPACITY		MANUFACTURER AND MODEL	REMARKS
								INPUT MBH	OUTPUT MBH		
<u>B-1(E)</u>	HOT WATER LOOP	98.3%	NATURAL GAS	100.0	150.0	40.0	3.5	1,000	982	LOCHINVAR FTX1000 (EXISTING)	1, 2
<u>B-2(E)</u>	HOT WATER LOOP	98.3%	NATURAL GAS	100.0	150.0	40.0	3.5	1,000	982	LOCHINVAR FTX1000 (EXISTING)	1, 2

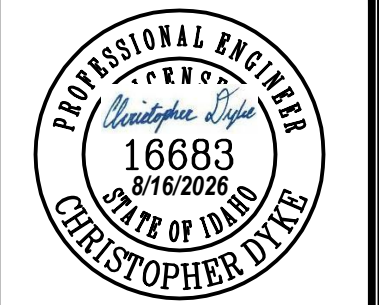
1. EXISTING BOILERS. SPECIFICATIONS SHOWN FOR REFERENCE ONLY.
2. SEE HEATING WATER SCHEMATIC FOR ADDITIONAL INFORMATION.
3. RECONFIGURE BOILER FROM VARIABLE PRIMARY TO PRIMARY-SECONDARY PUMPING ARRANGEMENT.

SYMBOL	SYSTEM LOCATION	SERVICE HORSEPOWER	V/Ø	PURPOSE	MANUFACTURER AND MODEL	REMARKS
<u>VFD-1</u>	MECHANICAL LOFT	2	460/3	<u>HWP-1</u>	ABB MODEL ACH 580	1, 2, 3, 4
<u>VFD-2</u>	MECHANICAL LOFT	2	460/3	<u>HWP-2</u>	ABB MODEL ACH 580	1, 2, 3, 4

1. APPROVED ALTERNATE MANUFACTURERS: DANFOSS, EATON, SQUARE D, AND YASKAWA.
2. PROVIDE W/PRESSURE SENSORS, INTERNAL FUSED DISCONNECT (W/ FUSES), NEMA 1 ENCLOSURE, WALL MOUNTING BRACKET, FACTORY AUTHORIZED START-UP, & 5% INTERNAL IMPEDANCE.
3. EQUIPMENT PROVIDED BY CONTROL CONTRACTOR, INSTALLED BY ELECTRICAL CONTRACTOR.
4. SEE CONTROL DRAWINGS FOR SEQUENCE OF OPERATION.

SYMBOL	AREA SERVED	TYPE	CAPACITY			MOTOR			OPERATING WEIGHT (LBS)	MANUFACTURER AND MODEL	REMARKS
			FLOW (GPM)	HEAD (FT)	MIN EFF	HP	RPM	V/Ø			
<u>HWP-1</u>	HOT WATER LOOP	INLINE VARIABLE SECONDARY	80	40	71	2	1,725	460/3	60	B&G E-90 2AB	1, 2, 3, 4
<u>HWP-2</u>	HOT WATER LOOP	INLINE VARIABLE SECONDARY	80	40	71	2	1,725	460/3	60	B&G E-90 2AB	1, 2, 3, 4
<u>BP-1</u>	BOILER PUMP (B-1)	INLINE PRIMARY	40	15	-	1/3	4,350	115/1	35	GRUNDFOS MAGNA3 40-80	1, 2, 3, 4
<u>BP-2</u>	BOILER PUMP (B-1)	INLINE PRIMARY	40	15	-	1/3	4,350	115/1	35	GRUNDFOS MAGNA3 40-80	1, 2, 3, 4
<u>CWP-1</u>	CHILLED WATER LOOP	INLINE PRIMARY	185	55	67	5	3325	460/3	85	B&G E-90 2AAB	1, 2, 3
<u>CWP-2</u>	CHILLED WATER LOOP	INLINE PRIMARY	185	55	67	5	3325	460/3	85	B&G E-90 2AAB	1, 2, 3

1. APPROVED ALTERNATE MANUFACTURERS: B&G, GRUNDFOS, TACO, PACO, & PATTERSON
2. PROVIDE UNIT WITH PREMIUM EFFICIENCY MOTOR.
3. CAPACITY BASED ON 70% WATER AND 30% PROPYLENE GLYCOL.
4. SEE CONTROL DRAWINGS FOR SEQUENCE OF OPERATION.

[illegible]

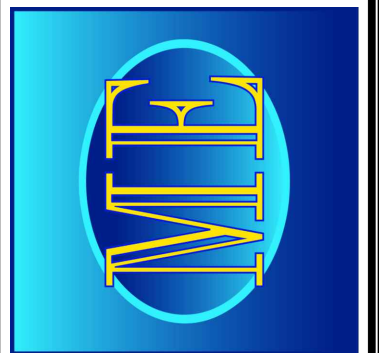
MUSGROVE
ENGINEERING, P.A.

234 S. Whisperwood Way
Boise, Idaho 83709
208-384-0585

1135 Pier View Drive
Idaho Falls, Idaho
208-523-2866

www.musgrovepa.com

OVER 50 YEARS OF EXCELLENCE



REBID - ITD DISTRICT 2
HEATING SYSTEM UPGRADES
2606 FRONTAGE RD
LEWISTON, ID 83501

PROJECT	26-150
DRAWN	JD
CHECKED	CD
DATE	AUGUST 2026
SHEET NAME	

MECHANICAL SCHEDULES

M3.0

GENERAL:
THE EXISTING CHILLED WATER SYSTEM SHALL CONSIST OF AN EXISTING OUTDOOR AIR-COOLED CHILLER, TWO (2) NEW CONSTANT SPEED PRIMARY CHILLED WATER PUMPS (LEAD / STANDBY OPERATION), AN EXISTING BUFFER TANK, AND AN EXISTING FLOW METER. THE CONTROL CONTRACTOR SHALL MODIFY THE EXISTING DDC CONTROL PACKAGE AS REQUIRED TO ACCOMMODATE THE NEW CONTROL SEQUENCE. THE NEW COMPONENTS SHALL BE INTEGRATED INTO THE EXISTING DELTA CONTROL SYSTEM.

TEXT IN GREY NOTES EXISTING CONTROL SEQUENCE. TEXT IN BOLD NOTES UPDATED CONTROL SEQUENCES..

OPERATION:
THE CHILLED WATER SYSTEM SHALL BE ENABLED WHENEVER ONE OF THE FOLLOWING CONDITIONS EXIST:

1. THE OUTSIDE AIR TEMPERATURE INCREASES TO THE COOLING ENABLE SET POINT FOR A PERIOD OF 10 CONSECUTIVE MINUTES.
2. THE CHILLED WATER BUFFER TANK TEMPERATURE RISES ABOVE 45°F FOR A PERIOD OF 30 SECONDS.
3. THERE IS A CALL FOR DEHUMIDIFICATION.

WHEN ONE OF THE ABOVE CONDITIONS IS MET, THE DDC CONTROLLER SHALL SEQUENCE THE FOLLOWING:

1. SEND AN **ENABLE COMMAND TO THE LEAD CHILLED WATER PUMP.**
 - a. VALIDATE THE RUNNING STATUS THROUGH THE CURRENT SENSING RELAY.
 - 1) IF THE LEAD PUMP FAILS TO START, THE STANDBY PUMP SHALL BE ENABLED.
 - 2) IF THE STANDBY PUMP FAILS TO START, AN ALARM OF THE EVENT SHALL BE SENT TO THE OPERATOR'S WORKSTATION.
2. SEND AN **ENABLE COMMAND TO THE AIR-COOLED CHILLER'S UNIT CONTROL PANEL (UCP).**
 - a. VALIDATE THE RUNNING STATUS OF THE CHILLER THROUGH THE UNIT CONTROL PANEL (UCP).
 - 1) IF THE CHILLER FAILS TO START, AN ALARM OF THE EVENT SHALL BE SENT TO THE OPERATOR'S WORKSTATION.

WHEN A RUNNING STATUS FROM THE CHILLER'S UNIT CONTROL PANEL (UCP) HAS BEEN CONFIRMED IT SHALL CONTINUE TO OPERATE UNDER ITS OWN INTERVAL CONTROLS AND MAINTAIN ITS OWN RAMPING AND LOADING RATES. THE DDC CONTROLLER SHALL SEND THE CHILLER'S UCP A CHILLED WATER SUPPLY TEMPERATURE SET POINT TO MAINTAIN (REFER TO THE CHILLED WATER SUPPLY TEMPERATURE RESET SCHEDULE).

THE COOLING ENABLE SET POINT SHALL BE SET AT 60°F.

THE CHILLED WATER SYSTEM SHALL BE DISABLED WHENEVER BOTH OF THE FOLLOWING CONDITIONS EXIST:

1. THE OUTSIDE AIR TEMPERATURE DECREASES 2" (ADJUSTABLE) OR MORE BELOW THE COOLING ENABLE SET POINT FOR A PERIOD OF 10 CONSECUTIVE MINUTES (ADJUSTABLE).
2. THE CHILLED WATER BUFFER TANK TEMPERATURE FALLS BELOW 45°F FOR A PERIOD OF 30 SECONDS.

WHEN ONE OF THE ABOVE CONDITIONS IS MET, THE DDC CONTROLLER SHALL SEQUENCE THE FOLLOWING:

1. SEND A **DISABLE COMMAND TO THE AIR-COOLED CHILLER'S UNIT CONTROL PANEL (UCP).**
 - a. VALIDATE THE RUNNING STATUS OF THE CHILLER THROUGH THE UNIT CONTROL PANEL (UCP).
 - 1) IF THE CHILLER FAILS TO STOP, AN ALARM OF THE EVENT SHALL BE SENT TO THE OPERATOR'S WORKSTATION.
2. SEND A **DISABLE COMMAND TO THE LEAD CHILLED WATER PUMP.**
 - a. WHEN THE CHILLED WATER SYSTEM IS DISABLED, THE LEAD PRIMARY CHILLED WATER PUMP SHALL CONTINUE TO OPERATE FOR A PERIOD OF 10 CONSECUTIVE MINUTES.
 - b. VALIDATE THE RUNNING STATUS THROUGH THE CURRENT SENSING RELAY.
 - 1) IF THE PUMP FAILS TO STOP, AN ALARM OF THE EVENT SHALL BE SENT TO THE OPERATOR'S WORKSTATION.

THE DDC CONTROLLER SHALL PROVIDE LEAD / STAND-BY CONTROL OF THE PRIMARY CHILLED WATER PUMPS. IT SHALL INCLUDE AN AUTOMATIC ROTATION OF THE LEAD PUMP BASED ON RUNTIME, FIXED ROTATION OR CALENDAR DATE. IN ADDITION, MANUAL OVERRIDE AND OPERATOR OVERRIDE CAPABILITIES SHALL ALSO BE PROVIDED.

COOLDOWN MODE OF OPERATION:

THE CHILLED WATER SYSTEM SHALL BE DISABLED 10 MINUTES PRIOR TO THE SWITCHOVER FROM THE OCCUPIED TO UNOCCUPIED MODE OF THE LAST AIR VALVE SYSTEM. THIS WILL ENABLE THE BUILDING TO UTILIZE THE RESIDUAL COOLING CAPACITY WITHIN THE CHILLED WATER PIPING SYSTEM.

CHILLED WATER SUPPLY TEMPERATURE SET POINT RESET SCHEDULE

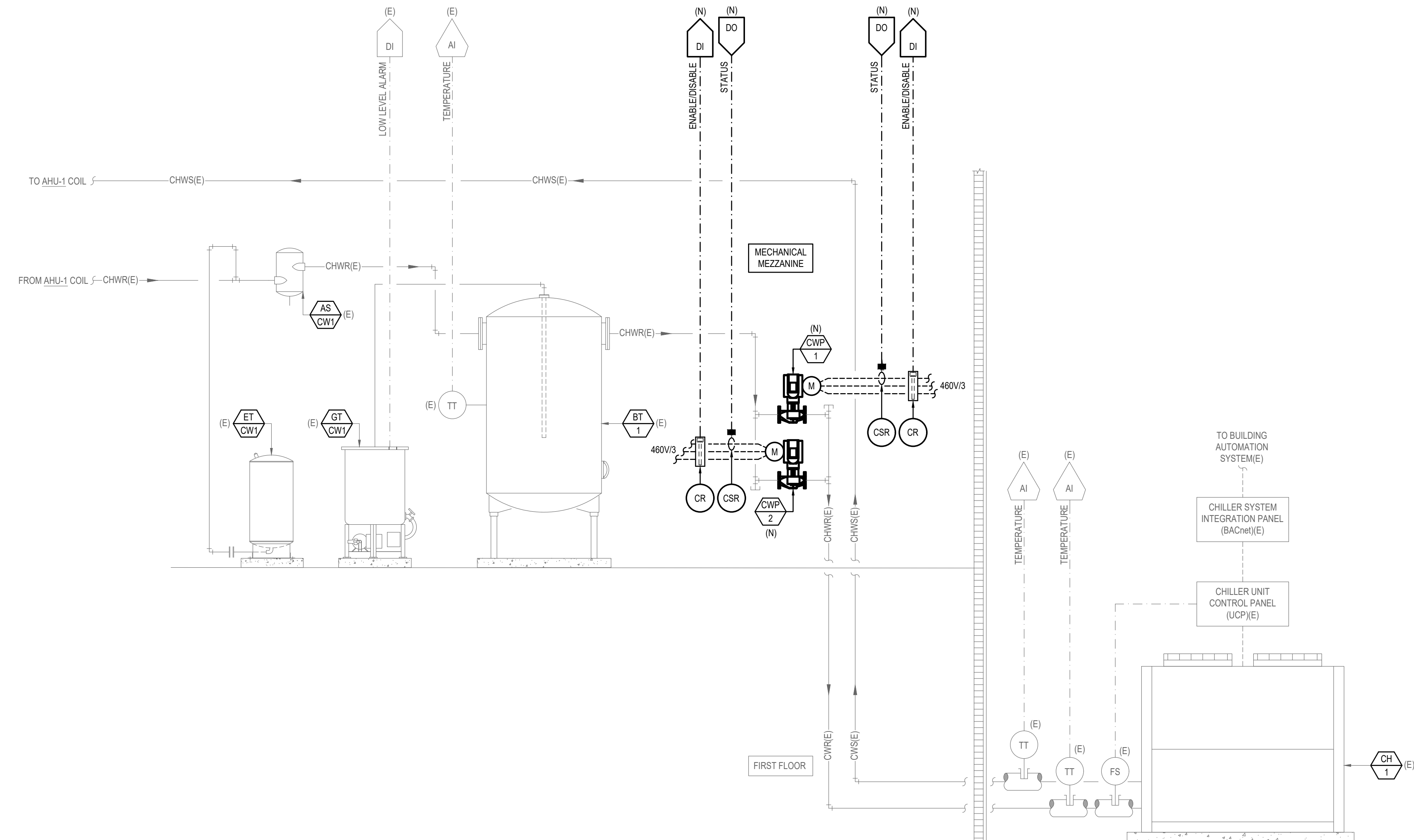
THE CHILLED WATER SUPPLY TEMPERATURE SET POINT SHALL BE RESET BY THE DDC CONTROLLER BASED ON OUTDOOR AIR TEMPERATURE. WHEN THE SYSTEM IS FIRST ENABLED, A CHILLED WATER SUPPLY TEMPERATURE OF 45°F SHALL BE PROVIDED AND SHALL DECREASE LINEARLY UP TO 55°F BASED ON THE FOLLOWING OUTDOOR AIR CONDITIONS:

1. OUTSIDE AIR TEMPERATURE = 60°F, SUPPLY WATER TEMPERATURE = 55°F
2. OUTSIDE AIR TEMPERATURE = 75°F, SUPPLY WATER TEMPERATURE = 45°F

CHILLER POINTS LIST:

ALL AVAILABLE CHILLER POINTS SUPPORTED BY THE CHILLER'S INTEGRATION MODULE SHALL BE MAPPED TO THE OPERATOR'S WORKSTATION. POINTS SHALL INCLUDE BUT NOT BE LIMITED TO THE FOLLOWING:

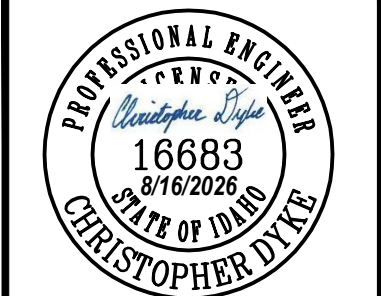
1. CHILLER ENABLE / DISABLE
2. CHILLER ENABLE / DISABLE STATUS
3. ENTERING CHILLED WATER TEMPERATURE
4. LEAVING CHILLED WATER TEMPERATURE
5. CHILLED WATER SET POINT
6. PERCENT TOTAL CAPACITY
7. ALARM STATE
8. CURRENT ALARMS
9. ALL COMPRESSOR CIRCUITS:
 - a. EVAPORATOR SATURATED REFRIGERANT PRESSURE
 - b. EVAPORATOR SATURATED REFRIGERANT TEMPERATURE
 - c. EXV POSITION



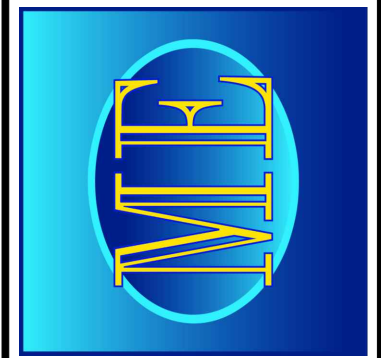
EXISTING CHILLED WATER SYSTEM SEQUENCE OF OPERATION

EXISTING CHILLED WATER SYSTEM CONTROL SCHEMATIC

NO.	REVISIONS					DATE



MUSGROVE
ENGINEERING, P.A.
234 S. Whisperwood Way
Bosc, Idaho 83709
208-384-0585
www.musgrovepa.com
OVER 50 YEARS OF EXCELLENCE



REBID - ITD DISTRICT 2
HEATING SYSTEM UPGRADES
2606 FRONTAGE RD
LEWISTON, ID 83501

PROJECT	26-150
DRAWN	JD
CHECKED	CD
DATE	AUGUST 2026
SHEET NAME	

MECHANICAL
CONTROLS

M4.1

A	AMPERES
AC	6" ABOVE BACKSPLASH
AC AFF	ABOVE FINISHED FLOOR
AFG	ABOVE FINISHED GRADE
AF	AMP FRAME
AIC	AMPS INTERRUPTING CAPACITY
AT	AMP TRIP
ATS	AUTOMATIC TRANSFER SWITCH
AWG	AMERICAN WIRE GAUGE
BD	BOTTOM OF DECK
BS	BOTTOM OF STRUCTURE
C	CEILING MOUNTED
C	CONDUIT
CB	CIRCUIT BREAKER
CF	COMPACT FLUORESCENT
CKT	CIRCUIT
CO	CONDUIT ONLY. PROVIDE PULL-LINE
CT	CURRENT TRANSFORMER
CTL	CONTROL
DC	DIRECT CURRENT
(D)	DEMOLITION ON
DEMO	DEMOLITION
DET	DETAIL
DTT	DOUBLE TWIN TUBE
E	EMERGENCY
(E)	EXISTING
EC	ELECTRICAL CONTRACTOR
EL	EMERGENCY LIGHT
F	FUSE
(F)	FUTURE
FACP	FIRE ALARM CONTROL PANEL
G/GND	GROUND
GFCI	GROUND FAULT CIRCUIT INTERRUPTER
GFI	GROUND FAULT INTERRUPTER
HH	HAND HOLE
HID	HIGH INTENSITY DISCHARGE
HOA	HAND-OFF-AUTO
HPS	HIGH PRESSURE SODIUM
HVAC	HEATING, VENTILATION, & AIR CONDITIONING
IG	ISOLATED GROUND
IPCO	IDAHO POWER COMPANY
J-BOX	JUNCTION BOX
KA	KILOAMP
KVA	KILO VOLT-AMP
KW	KILOWATT
KWH	KILOWATT HOUR
LCP	LIGHTING CONTROL PANEL
MB	MAIN BREAKER
MBR	MAIN CIRCUIT BREAKER
MCC	MOTOR CONTROL CENTER
MDP	MAIN DISTRIBUTION PANEL
MLO	MAIN LUGS ONLY
MMC	MODULAR METERING CENTER
MH	METAL HALIDE
MSB	MAIN SWITCH BOARD
MTG	MOUNTING
N	NEUTRAL
(N)	NEW
NC	NORMALLY CLOSED
NEC	NATIONAL ELECTRICAL CODE
NIC	NOT IN CONTRACT
NL	NIGHT LIGHT
NO	NORMALLY OPEN
NTS	NOT TO SCALE
OH	OVERHEAD
OS	OCCUPANCY SENSOR
P	POLES
PC	PHOTO-CONTROL
PVC	POLYVINYL CHLORIDE
PWR	POWER
RE-	REFERENCE
REC	RECEPTACLE
(R)	RELOCATED
SF	SQUARE FEET
TBD	TO BE DETERMINED
TDR	TIME DELAY RELAY
TK	TOE KICK
TS	TWISTED SHIELDED PAIR
TRT	TRIPLE TUBE
TTB	TELEPHONE TERMINAL BOARD
(TYP.)	TYPICAL
UC	UNDERCABINET
UG	UNDERGROUND
U.N.O.	UNLESS NOTED OTHERWISE
V	VOLT
V	VOLT-AMPERE
W	WATT
WG	WIRE GUARD
WP	WEATHER PROOF/NEMA 3R
PROVIDED/ PROVIDE AND INSTALL/ INSTALL	PROVIDE AND INSTALL / PROVIDED AND INSTALLED BY / PROVIDE AND INSTALL
NOTE:	THIS IS A STANDARD LIST OF COMMONLY USED ELECTRICAL ABBREVIATIONS. SOME OF THE ABBREVIATIONS SHOWN ABOVE MAY NOT BE USED IN THIS DRAWING PACKAGE.

REFERENCE FIXTURE SCHEDULE FOR MOUNTING TYPE, MOUNTING HEIGHT, AND FIXTURE TYPE.	
	DOUBLE FACE EXIT SIGN, CEILING MOUNTED, PROVIDE UNSWITCHED CONDUCTOR.
	WALL MOUNTED DOUBLE FACE EXIT SIGN PROVIDE UNSWITCHED CONDUCTOR. MOUNT AT +8'-0" UNO.
	SINGLE FACE EXIT SIGN, CEILING MOUNTED PROVIDE UNSWITCHED CONDUCTOR.
	WALL MOUNTED SINGLE FACE EXIT SIGN PROVIDE UNSWITCHED CONDUCTOR. MOUNT AT +8'-0" UNO.
	ARROW INDICATES DIRECTION TO BE SHOWN ON SIGN.
	1'X1' LIGHT FIXTURE.
	1'X1' LIGHT FIXTURE, PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.
	1'X4' LIGHT FIXTURE.
	1'X4' LIGHT FIXTURE, PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.
	2'X4' LIGHT FIXTURE.
	2'X4' LIGHT FIXTURE, PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.
	2'X2' LIGHT FIXTURE.
	2'X2' LIGHT FIXTURE, PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.
	DIRECT/INDIRECT LIGHT FIXTURE. SEE SCHEDULE FOR LENGTH.
	DIRECT/INDIRECT LIGHT FIXTURE. SEE SCHEDULE FOR LENGTH. PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR
	STRIP LIGHT FIXTURE. SEE SCHEDULE FOR LENGTH.
	STRIP LIGHT FIXTURE. SEE SCHEDULE FOR LENGTH. PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR
	WALL MOUNTED LIGHT FIXTURE.
	WALL MOUNTED LIGHT FIXTURE, PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.
	RECESSED LIGHT FIXTURE
	RECESSED LIGHT FIXTURE. PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.
	ROUND LIGHT FIXTURE
	ROUND EMERGENCY LIGHT FIXTURE. PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.
	WALL MOUNTED LIGHT FIXTURE.
	WALL MOUNTED EMERGENCY LIGHT FIXTURE. PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR.
	POLE LIGHT 1 HEAD WITH POLE
	TIME CLOCK
	PHOTO CONTROL CELL LOCATED 12" ABOVE ROOF FACING NORTH.
	OCCUPANCY SENSOR. PROVIDE RELAYS AND POWER PACKS AS REQUIRED.
	EMERGENCY EGRESS LIGHTING WITH OUT FIXTURE HEADS. CONNECT TO AN UNSWITCHED CONDUCTOR.
	EMERGENCY EGRESS LIGHTING. CONNECT TO AN UNSWITCHED CONDUCTOR.
	WALL MOUNTED SINGLE FACE EXIT SIGN WITH EMERGENCY EGRESS LIGHTING. PROVIDE UNSWITCHED CONDUCTOR. MOUNT AT +8'-0" UNO.
	CEILING MOUNTED, SINGLE FACE EXIT SIGN WITH EMERGENCY EGRESS LIGHTING. PROVIDE UNSWITCHED CONDUCTOR.
	CEILING MOUNTED, DOUBLE FACE EXIT SIGN WITH EMERGENCY EGRESS LIGHTING. PROVIDE UNSWITCHED CONDUCTOR.
XXX	INDICATES FIXTURE TYPE. REFER TO FIXTURE SCHEDULE.
	EXTERIOR WALL PACK
	EMERGENCY EXTERIOR WALL PACK. PROVIDE EMERGENCY BATTERY BACKUP CONNECTED TO AN UNSWITCHED CONDUCTOR

SW	SWITCH, TYPE AS INDICATED. +46" AFF
2	DOUBLE POLE
3	3-WAY
4	4-WAY
K	KEYED
P	PILOT LIGHT
D	DIMMER
HP	HORSEPOWER RATED
TO	THERMAL OVERLOAD
LV	LOW VOLTAGE
OS	OCCUPANCY SENSOR
OR	LOW VOLTAGE, MOMENTARY OVERRIDE
VS	VACUANCY SENSOR
a	SUPERSCRIPT INDICATES LIGHTS TO BE SWITCHED TOGETHER
SS	DUAL LEVEL SWITCHING, INSIDE AND OUTSIDE LAMPS OF FIXTURE TO BE SWITCHED SEPARATELY.
SS _o	DUAL LEVEL SWITCHING WITH OCCUPANCY SENSOR, INSIDE AND OUTSIDE LAMPS OF FIXTURE TO BE SWITCHED SEPARATELY.
SS _o	OCCUPANCY SENSOR WITH MANUAL DIMMING, SET FOR 50% AUTOMATIC ON, AUTOMATIC OFF, WITH MANUAL DIMMING.
	SINGLE CONVENIENCE OUTLET, +18" AFF UNO
	FLOOR MOUNT SINGLE CONVENIENCE OUTLET
	DUPLEX CONVENIENCE OUTLET, +18" AFF UNO
	FLOOR MOUNT DUPLEX CONVENIENCE OUTLET
	EMERGENCY DUPLEX CONVENIENCE OUTLET, +18" AFF UNO
	SWITCHED DUPLEX CONVENIENCE OUTLET, +18" AFF UNO
	FLOOR MOUNTED SWITCHED DUPLEX CONVENIENCE OUTLET
	FOURPLEX CONVENIENCE OUTLET. +18" AFF UNO
	FLOOR MOUNT FOURPLEX CONVENIENCE OUTLET
	CONNECTION POINT TO EQUIPMENT SPECIFIED. ELECTRICAL CONTRACTOR TO SUPPLY RACEWAY AND CONDUCTORS AND MAKE FINAL CONNECTION TO EQUIPMENT UNDER THIS SECTION. UNO
	FLOOR MOUNTED CONNECTION POINT, SEE NOTE ABOVE FOR REQUIREMENTS
	FLOOR MOUNTED JUNCTION BOX
	JUNCTION BOX
	WALL MOUNTED PUSH BUTTON, MOUNT AT SWITCH HEIGHT UNO
	WALL MOUNTED PUSH BUTTON, HANDICAPPED MOUNT AT SWITCH HEIGHT UNO
	WALL MOUNTED PUSH BUTTON, MOUNT AT SWITCH HEIGHT UNO
	MOTOR STARTER/CONTACTOR, SIZE/POLES NEMA 1 UNO AS INDICATED
	COMBINATION STARTER AND DISCONNECT, SIZE/POLES, STARTER SIZE AS INDICATED, NEMA 1 UNO
	FUSED DISCONNECT SWITCH, SIZE/POLES, FUSE SIZES AS INDICATED, NEMA 1 UNO
	NON-FUSED DISCONNECT SIZE/ POLES AS INDICATED, NEMA 1 UNO
	THERMOSTAT, +46" AFF PROVIDE CONDUIT, J-BOX, CONDUCTORS AS REQUIRED TO CONTROL ASSOCIATED UNITS. UNO COORDINATE WITH DIVISION 15.
	HUMIDISTAT, +46" AFF PROVIDE CONDUIT, J-BOX, CONDUCTORS AS REQUIRED TO CONTROL ASSOCIATED UNITS.
	POWER POLE - DUAL CHANNEL
	RECESSED ENTERTAINMENT BOX
	TRANSFORMER
	PANELBOARD. SEE SCHEDULE FOR TYPE.
	EQUIPMENT CABINET, SURFACE MOUNTED
	EQUIPMENT CABINET FLUSH MOUNTED
	SURFACE MULTI-OUTLET RACEWAY
	MECHANICAL EQUIPMENT CALL OUT
	KITCHEN EQUIPMENT CALLOUT

A. THESE ELECTRICAL DRAWINGS ARE DIAGRAMMATIC IN NATURE; THEREFORE, THE ELECTRICAL CONTRACTOR SHALL COORDINATE ALL ELECTRICAL EQUIPMENT AND DEVICE LOCATIONS WITH ARCHITECTURAL, MECHANICAL, AND PLUMBING DIVISIONS PRIOR TO ROUGH-IN. REFER TO AND COORDINATE WITH ARCHITECTURAL, MECHANICAL, AND PLUMBING DRAWINGS FOR ADDITIONAL WORK THAT IS REQUIRED BY THE ELECTRICAL CONTRACTOR.

B. ALL CONDUIT AND JUNCTION BOXES ARE TO BE CONCEALED UNLESS LOCATED WITHIN DEDICATED ELECTRICAL OR MECHANICAL ROOMS. USE OF SURFACE MOUNTED RACEWAYS IN ALL OTHER SPACES MUST BE APPROVED BY THE ARCHITECT FOR EACH LOCATION. WHERE SURFACE RACEWAYS ARE APPROVED, UTILIZE WIREMOLD, OR APPROVED EQLANT, SURFACE MOUNTED RACEWAYS PAINTED TO MATCH SURROUNDING WALLS.

C. REFER TO ARCHITECTURAL ELEVATIONS FOR OUTLET HEIGHTS WHERE THE SPECIFIC OUTLET HEIGHT IS NOT INDICATED. REFER TO THE ELECTRICAL LEGEND FOR THE DEFAULT OUTLET HEIGHT WHEN NOT INDICATED ON ELEVATIONS OR ON AT THE DEVICES.

D. PROVIDE PULL-LINE IN ALL EMPTY CONDUITS.

E. TERMINATE ALL LOW-VOLTAGE CONDUITS WITH INSULATED THROAT BUSHING.

F. MECHANICAL EQUIPMENT INDICATED IS SHOWN IN AN APPROXIMATE LOCATION. COORDINATE EXACT LOCATION WITH MECHANICAL CONTRACTOR PRIOR TO ROUGH-IN.

FIRE ALARM:

G. INSTALL PLENUM RATED FIRE ALARM CONDUCTORS FROM ALL FIRE ALARM DEVICES INDICATED TO THE FIRE ALARM CONTROL PANEL OR NAC EXTENDER PANEL(S) AS REQUIRED. STUB 3/4" CONDUIT FROM DEVICE TO VOID ABOVE CEILING. PROVIDE NAC EXTENDER PANEL(S) QUANTITY AS REQUIRED. IN LOCATIONS INDICATED AND CIRCUITING AS REQUIRED FOR A COMPLETE INSTALLATION. CIRCUIT THE FIRE ALARM NOTIFICATION AND INITIATION DEVICES PER THE ELECTRICAL SPECIFICATIONS. FURNISH AND INSTALL ALL APPURTENANCES AND PROGRAMMING REQUIRED FOR A COMPLETE AND OPERATIONAL SYSTEM. REFER TO ELECTRICAL FIRE ALARM SPECIFICATIONS FOR SYSTEM REQUIREMENTS AND SUBMITTAL PROCEDURES.

SITE:

H. CONTRACTOR SHALL COORDINATE WITH AN UNDERGROUND LOCATING SERVICE PRIOR TO COMMENCING WORK. SEE CIL DRAWINGS FOR ADDITIONAL SITE INFORMATION. COORDINATE WITH OTHER SITE DISCIPLINES.

I. SITE LIGHTING AND UTILITY EQUIPMENT SHOWN IN APPROXIMATE LOCATION. COORDINATE EXACT LOCATION WITH CIL DRAWINGS, PROPERTY LINES, AND UTILITY COMPANIES PRIOR TO ROUGH-IN.

J. REFER TO POLE BASE DETAIL FOR SITE LIGHTING POLE BASE REQUIREMENTS.

K. ROUTE CONDUITS IN COMMON TRENCH WHEN POSSIBLE REFER TO TRENCHING DETAIL.

DEMOL:

L. THE ELECTRICAL DEMOLITION DRAWING(S) PROVIDED ARE INTENDED TO ASSIST THE ELECTRICAL CONTRACTOR IN ESTABLISHING AREAS REQUIRING DISCONNECTION, REMOVAL, OR RELOCATION OF ELECTRICAL EQUIPMENT, OUTLETS, WIRING, DEVICES, FIXTURES, ETC., AND MAY NOT INDICATE ALL DEVICES OR THE FULL EXTENT OF DEMOLITION AND RECONNECTION WHICH MAY BE REQUIRED. THE ELECTRICAL CONTRACTOR SHALL VISIT THE JOB SITE AND THOROUGHLY EXAMINE ALL REQUIRED DEMOLITION WORK AND INCLUDE ALL LABOR AND INCIDENTALS THAT WILL BE NECESSARY TO PERFORM DEMOLITION RECONNECTION AND TEMPORARY POWER CONNECTIONS IN THE BID.

M. ALL ELECTRICAL DEVICES AND WALLS INDICATED ON THE ELECTRICAL DEMOLITION DRAWING(S) ARE TO REMAIN UNLESS OTHERWISE NOTED.

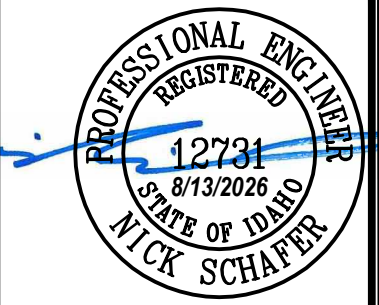
PANEL:		LB	PROJECT:		ITD MATERIALS LAB		AMPERE RATING:		400 MLO	SC RATING:		22000 A/C	
VOLTAGE:		120/240	1	PH	3	WIRE	REMARKS:		EXISTING PANEL TO REMAIN		MOUNTING:		22000 A/C SURFACE
OCT NOTES:													
1 RED HAND LOCKABLE BREAKER													
2 GFCI FOR PERSONNEL PROTECTION (5mA)													
3 GFCI FOR EQUIPMENT PROTECTION (30mA)													
4 AFC COMBINATION TYPE BREAKER													
OCT	DESCRIPTION	OCT NOTE	LOAD AMPS	AMPS/POLES	A	PHASE (AMPS)	C	AMPS/POLES	LOAD AMPS	OCT NOTE	DESCRIPTION	OCT	
2	REC- ASHLT LAB NW COUNTER NW	5	1.5	20	1	19.5		30	2	18.0	5	GRIEVE OVEN ASHPALT LAB E	2
3	NCAAT ASHPALT LAB NW	5	27.0	35	2		45	**	18.0	5	***		4
5	***	5	27.0	**	**	45		30	2	18.0	5	GRIEVE OVEN ASHPALT LAB E	6
7	NCAAT ASHPALT LAB NW	5	27.0	35	2		45	**	18.0	5	***		10
9	***	5	27.0	**	**	54		35	2	27.0	5	GRIEVE OVEN ASHPALT LAB E	11
11	GRIEVE OVEN ASHPALT LAB NW	5	18.0	30	2		45	**	27.0	5	***		12
13	***	5	18.0	**	**	45		35	2	27.0	5	GRIEVE OVEN ASHPALT LAB E	14
15	GRIEVE OVEN ASHPALT LAB NW	5	18.0	30	2		45	**	27.0	5	***		16
17	***	5	18.0	**	**	45		20	2	18.0	5	REC- ASHPALT LAB E N WALL	17
19	WVR OVEN ASHPALT LAB NW	5	12.0	20	1	19.5	13.5	20	1	1.5	5	REC- ASHPALT LAB E N WALL	20
21	WVR OVEN ASHPALT LAB NW	5	12.0	20	1	28		30	2	16.0	5	HAMBURG WHEEL TRACKER	22
23	REC- ASHPALT LAB NW E WALL	5	1.5	20	1		17.5	**	16.0	5	***		24
25	REC- ASHPALT LAB NW CEILING	5	1.5	20	1	17.5		30	2	16.0	5	***	26
27	REC- ASHPALT LAB NW CEILING	5	1.5	20	1		17.5	**	16.0	5	***	28	
29	REC- ASHPALT LAB NW CEILING	5	1.5	20	1	3		20	1	1.5	5	REC- PERFORM LAB W WALL	30
31	REC- ASHPALT LAB NW S WALL	5	3.0	20	1		8	20	1	5.0	5	REC- ROTARY EVAPORATOR	32
33	REC- ASHPALT LAB NW S WALL	5	3.0	20	1	41		20	1	1.5	5	REC- PERFORM LAB W WALL	34
35	GRIEVE OVEN PERFORM LAB	5	18.0	30	2		23	20	1	3.0	5	REC- ROTARY EVAPORATOR	36
37	***	5	18.0	**	**	25		20	1	3.0	5	REC- PERFORM LAB S WALL	38
39	GRIEVE OVEN PERFORM LAB	5	18.0	30	2		21	20	1	3.0	5	REC- PERFORM LAB E WALL	40
41	***	5	18.0	**	**	23.8		20	1	5.8	5	REC- FORNEY COMPRESSION MACH	42
43	LA ABRASION MACHINE	5	5.8	20	2		7.1	20	1	1.5	5	REC- CEILING PERFORM LAB	44
45	***	5	5.8	**	**	7.3		20	1	1.5	5	REC- CEILING PERFORM LAB	46
47	REC- 230V STORAGE	5	5.0	30	2		12.5	20	1	7.5	5	REC- STORAGE	48
49	***	5	5.0	**	**	10		30	2	5.0	5	REC- 230V STORAGE	50
51	REC- MECH LOFT	5	7.5	20	1		12.5	**	5.0	5	***	52	
53	EF7	5	9.8	20	1	14.4		3	4.8	6	GF4	54	
55	LOUVERS L1, PH-1	5	1.5	20	1	7.6	6.1	**	4.8	5	***	56	
57	BOILERS 1 AND 2	3	3.0	20	1			**	4.8	5	***	58	
59	EH-4	5	8.3	20	2		15.5	20	1	7.2	5	GT-HW1	60
61	***	5	6.3	**	**	15.5		20	1	7.2	5	GT-HW1	62
63	GENERATOR HEATER	5	3.0	20	1			20	3.0	5	DRINKING FOUNTAIN	64	
65	GENERATOR BATTERY	5	3.0	20	1	6		20	1	3.0	5	REC- AIR COMP ACCESSORIES	66
67	HEAT TAPE	5	5.0	20	1		12.2	15	1	7.2		BP-1	68
69	HUMIDIFIER 1	5	4.5	20	1	11.7		15	1	7.2		BP-2	70
71	SV AND EV IN LOFT	5	3.0	20	1		3	20	1		5	SPARE	72
73	SV AND EV IN LABS	5	3.0	20	1	3		20	1		5	SPARE	74
75	SPARE	5	5	20	1		0	20	1		5	SPARE	76
77	SPARE	5	5	20	1	0		0	20	1	5	SPARE	78
79	SPARE	5	5	20	1		0	20	1		5	SPARE	80
81	RESERVED					61.2		200	2	61.2	5	S (PANEL EM)	82
83	RESERVED						48.8	**	48.8	5	***		84
						417.5		404.4	AMPS				
						50.1		48.5	KW	98.6	TOTAL KW		

PANEL: HM			PROJECT: ITD MATERIALS LAB								
VOLTAGE: 480/277			3 PH	4 WIRE	AMPERE RATING:	200 MLO	SC RATING:	35000 AIC			
CKT NOTES:						REMARKS:					
1. RED HANDLE LOCKABLE BREAKER			5. EXISTING BREAKER			EXISTING PANEL TO REMAIN					
2. GFCI FOR PERSONNEL PROTECTION (5mA)											
3. GFCI FOR EQUIPMENT PROTECTION (30mA)											
4. AFCI COMBINATION STYLE BREAKER											
CKT	DESCRIPTION	CKT NOTE	LOAD POLES	AMPS/ POLES	PHASE (AMPS) A B C	AMPS/ POLES	LOAD POLES	CKT NOTE	DESCRIPTION	CKT	
1	CWP-1		5	7.6 15 3	15.2	15 3	7.6 5		CWP-2	2	
2			5	7.6 15 3			7.6 5			4	
5	***		5	7.6 15 3		15.2	7.6 5			6	
7	HWP-1, HWP-2		6.0 15 3	12			6.0		VFD-1, VFD-2	8	
9	***		6.0 15 3		12		6.0		***	10	
11	***		6.0 15 3			12	6.0		***	12	
13	SPARE		5	15 3 7.6			15 3	7.6 5	EUF-1	14	
15	***		5	15 3 7.6	7.6		15 3	7.6 5	***	16	
17	***		5	15 3 7.6		7.6	15 3	7.6 5	***	18	
19	AHU-1		5	32.0 60 3	35		15 3	3.0 5	EUF-2	20	
21	***		5	32.0 60 3	35		15 3	3.0 5	***	22	
23	***		5	32.0 60 3		35	15 3	3.0 5	***	24	
25	EUF-5		5	3.0 15 3	7.8		15 3	4.8 5	EUF-3	26	
27	***		5	3.0 15 3	7.8	7.8	15 3	4.8 5	***	28	
29	***		5	3.0 15 3		7.8	15 3	4.8 5	***	30	
31	SPARE		20	1 0			20 1		SPARE	32	
33	SPARE		20	1 0			20 1		SPARE	34	
35	SPARE		20	1 0			20 1		SPARE	36	
37	SPARE		20	1 0			20 1		SPARE	38	
39	SPARE		20	1 0			20 1		SPARE	40	
41	SPARE		20	1 0			20 1		SPARE	42	
					77.6 77.6 77.6	AMPS					
					21.5 21.5 21.5	KW	64.5 TOTAL KW				

The diagram illustrates the marking and identification of conductors and conduits according to the National Electrical Code (NEC). It shows various components and their labeling requirements:

- DESIGNATES CIRCUIT ON EMERGENCY SOURCE:** Points to a marking on the conduit.
- UNMARKED CIRCUIT IS CONCEALED IN CEILING OR WALL. MAINTAIN CONDUIT AND CONDUCTOR SIZE THROUGHOUT ENTIRE CIRCUIT.** Points to a section of conduit.
- BEGINNING OF INDIVIDUAL CIRCUIT(S). CIRCUIT NUMBER(S) INDICATED.** Points to a marking on the conduit.
- CONDUIT DOWN / CONDUIT UP:** Points to arrows indicating the direction of conduit run.
- EXISTING:** Points to a dashed line representing an existing conduit.
- CONCEALED IN FLOOR OR UNDERGROUND:** Points to a section of conduit.
- CONDUIT, STUBBED, CAPPED AND MARKED WITH PULL CORD AS SPECIFIED:** Points to a marking on the conduit.
- CURRENT CARRYING CONDUCTORS:** Points to a conductor.
- NEUTRAL CONDUCTORS:** Points to a conductor.
- GROUNDING CONDUCTOR:** Points to a conductor.
- PANEL NAME:** Points to a marking on the panel.
- CIRCUIT #:** Points to a marking on the panel.
- PANEL HOMERUN (3/4"-2#12, 1#12G CONDUCTORS UNO):** Points to a conductor.
- RACEWAY:** Points to a raceway.
- 1"-4#10, 1#10G:** Points to a conductor.
- SIZE:** Points to the conductor size.
- QUANTITY:** Points to the conductor quantity.

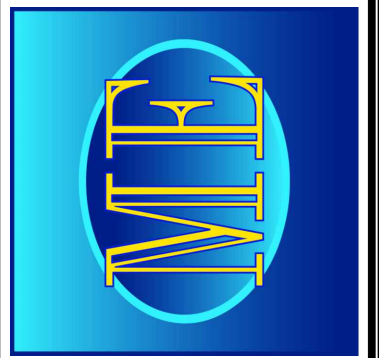
NOTE:
EDISON STYLE SHARED NEUTRAL CONDUCTORS ARE NOT ALLOWED. EACH 1 POLE BREAKER SHALL BE FURNISHED WITH AN INDIVIDUAL DEDICATED NEUTRAL CONDUCTOR.



MUSGROVE
ENGINEERING, P.A.

2234 S. Whisperwood Way
Boise, Idaho 83709
208-384-0585

www.musgrovepa.com
OVER 50 YEARS OF EXCELLENCE



REBID - ITD DISTRICT 2
HEATING SYSTEM UPGRADES
2606 FRONTAGE RD
LEWISTON, ID 83501

PROJECT 26-150

DRAWN JRR

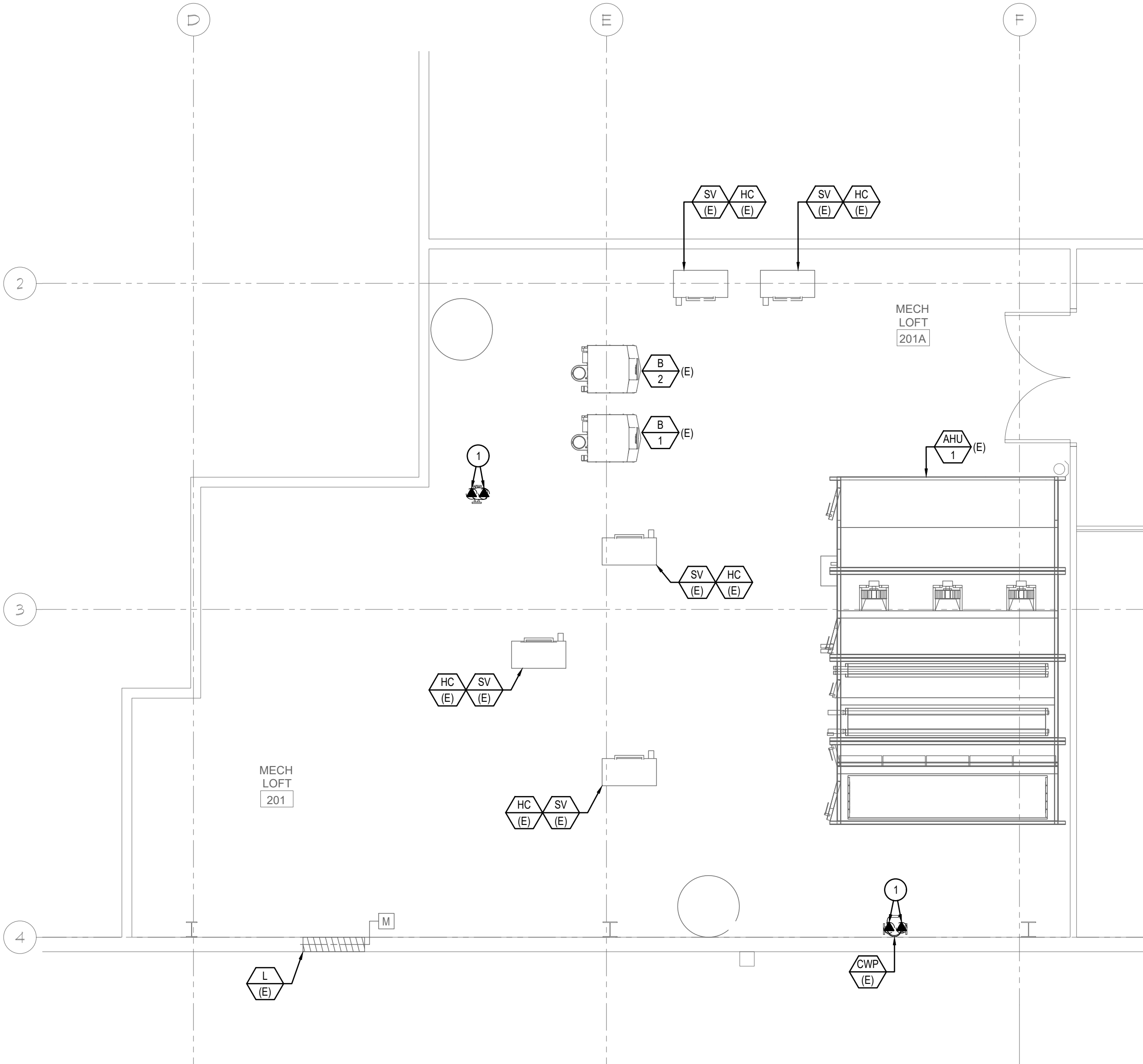
CHECKED_{NS}

DATE AUGUST 202

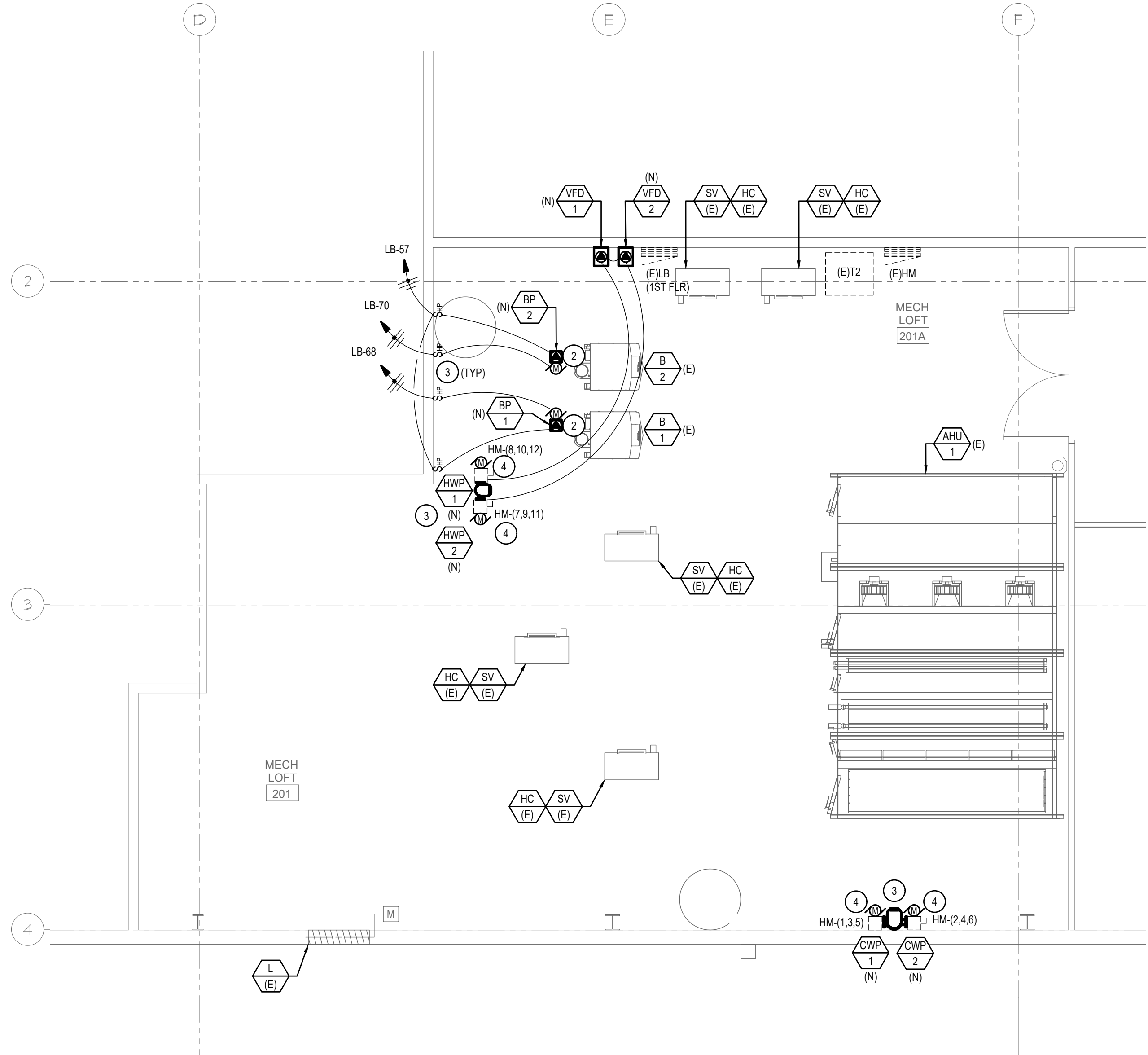
SHEET NAME

ELECTRICAL COVER SHEET

EO.0



1 ELECTRICAL LOFT DEMO PLAN
1/4" = 1'-0"
NORTH



2 ELECTRICAL LOFT NEW PLAN
1/4" = 1'-0"
NORTH

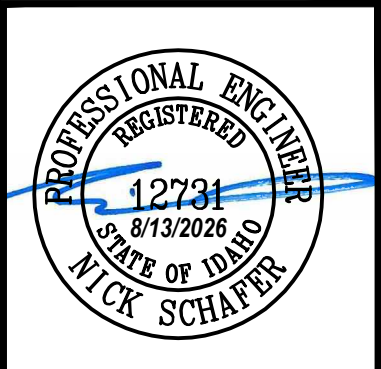
KEYED NOTES:

- # SYMBOL USED FOR NOTE CALLOUT.
- EXISTING DEVICE TO BE DEMOLISHED. COORDINATE DISCONNECTION WITH MECHANICAL CONTRACTOR. DEMOLISH AND REMOVE CONDUIT AND CONDUCTORS BACK TO SOURCE. EXISTING RACEWAYS MAY BE REUSED IF SUITABLE FOR REUSE.
 - 1/2" CONDUIT TO CORRESPONDING BOILER FOR INTERLOCK. BOX, CONDUIT, NYLON PULLSTRING TO BE PROVIDED BY ELECTRICAL CONTRACTOR. MECHANICAL CONTRACTOR TO MAKE FINAL CONNECTIONS. COORDINATE BOX SIZE AND QUANTITY AND EXACT LOCATION WITH MECHANICAL CONTRACTOR.
 - FIELD COORDINATE DISCONNECT AND EQUIPMENT LOCATION WITH MECHANICAL CONTRACTOR TO MAINTAIN ALL REQUIRED CLEARANCES.
 - NEW EQUIPMENT TO REPLACE EXISTING 1.1. INTERCEPT EXISTING CONDUIT AND CONDUCTORS AND EXTEND TO NEW REPLACEMENT EQUIPMENT. PROVIDE MATERIALS AS NECESSARY FOR EQUIPMENT OPERATION. UTILIZE EXISTING DISCONNECT TO MAKE CONNECTIONS TO NEW PUMP.

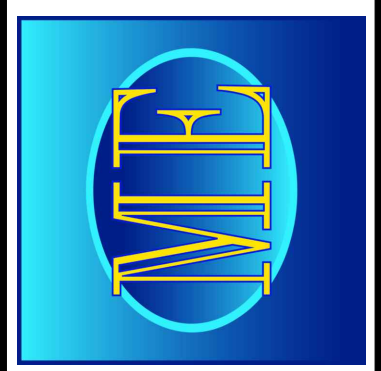
GENERAL NOTES:

- MECHANICAL EQUIPMENT SHOWN IN APPROXIMATE LOCATION. COORDINATE WITH MECHANICAL CONTRACTOR.
- THESE ELECTRICAL DRAWINGS ARE DIAGRAMMATIC IN NATURE; THEREFORE, THE ELECTRICAL CONTRACTOR SHALL COORDINATE ALL ELECTRICAL EQUIPMENT AND DEVICE LOCATIONS WITH ARCHITECTURAL, MECHANICAL, AND PLUMBING DIVISIONS PRIOR TO ROUGH-IN. REFER TO AND COORDINATE WITH ARCHITECTURAL, MECHANICAL, AND PLUMBING DRAWINGS FOR ADDITIONAL WORK THAT IS REQUIRED BY THE CONTRACTOR.
- ALL CONDUIT AND JUNCTION BOXES ARE TO BE CONCEALED IN NEW WALLS, EXISTING FURRED OUT WALLS AND EXISTING ACCESSIBLE CEILINGS. USE OF SURFACE MOUNTED RACEWAYS MUST BE APPROVED BY THE ARCHITECT FOR EACH LOCATION. WHERE APPROVED UTILIZE WIREMOLD OR APPROVED EQUAL SURFACE MOUNTED RACEWAYS PAINTED TO MATCH SURROUNDING WALLS.

NO.	REVISIONS	DATE



MUSGROVE
ENGINEERING, P.A.
234 S. Whisperwood Way
Boise, Idaho 83709
208-384-0585
www.musgrovepa.com
208-523-2862
OVER 50 YEARS OF EXCELLENCE



REBID - ITD DISTRICT 2
HEATING SYSTEM UPGRADES
2606 FRONTAGE RD
LEWISTON, ID 83501

PROJECT	26-150
DRAWN	JRR
CHECKED	NS
DATE	AUGUST 2026
SHEET NAME	

MECHANICAL
POWER PLAN

E1.0